

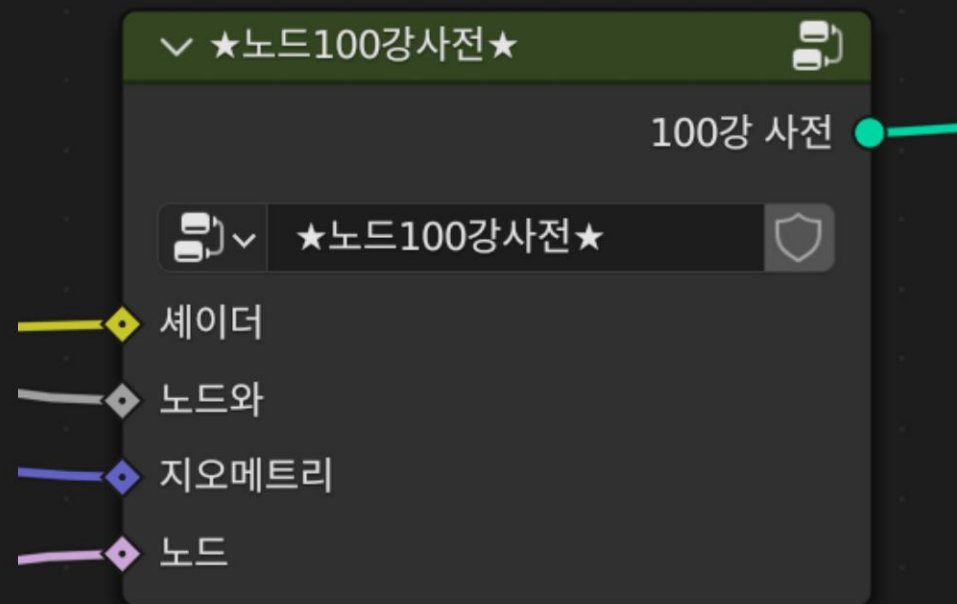
001 WELCOME

Node basic concept

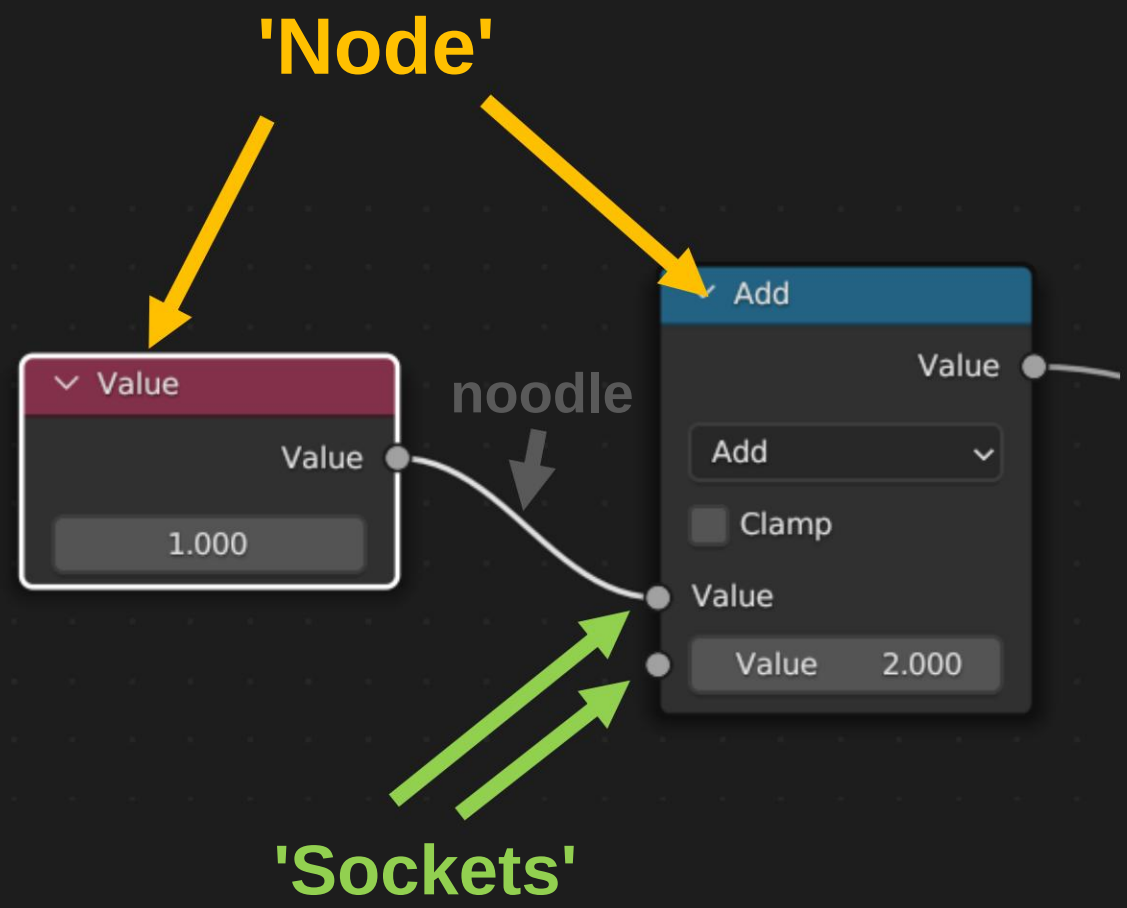
Things to prepare before listening to

the lecture - Workspace

setting - Explanation of lecture materials



Node?



What this class covers

Understanding and application of shader nodes and geometry nodes

Shader Nodes



Geometry Nodes



Putting it all together...

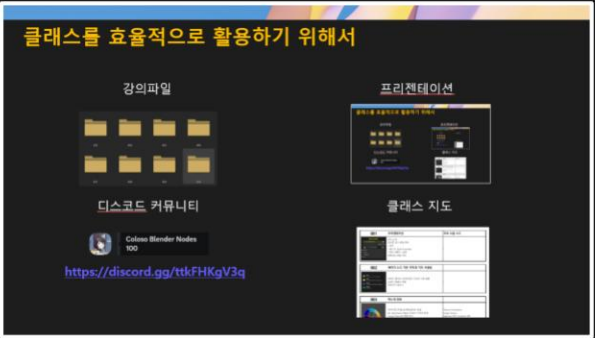


In order to use the class effectively

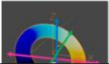
lecture file (.BLEND, image..)



LecturePDF



class map

001	오리엔테이션	주요 사용 노트
	강의 소개 강의를 듣기 위한 준비 - 사전 - 예제 문, Quick Favorites - 워크스페이스 설정 제공되는 파일 안내	
002	셰이더 노트 기본 조작과 기초 사용법	
	셰이더 에디터, 마티컬의 구조와 기본 흐름 소재의 종류와 연결 여러가지 셰이더	
003	텍스처 좌표	
	이미지를 3차원 표면에 입히는 방법 UV, Generated, Object 좌표의 의미와 특징 Image Texture의 사용 방법	Texture Coordinate Image Texture Separate XYZ, Combine XYZ

Discord Community



Coloso Blender Nodes
100

<https://discord.gg/ttkFHKgV3q>

Preparation (1)

Specifications of the computer in which the lecture is being produced

I7 4790K / RTX3070 8GB / 16G DDR4 / WINDOWS 10

version

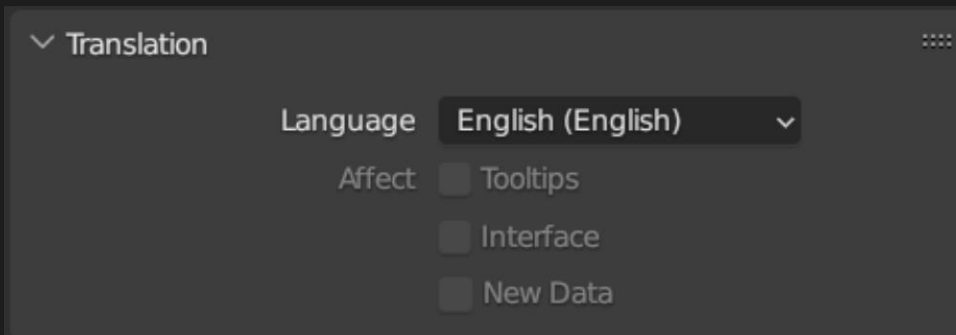


Blender 3.5 (or higher)

Simulation node only progressed to 3.6.

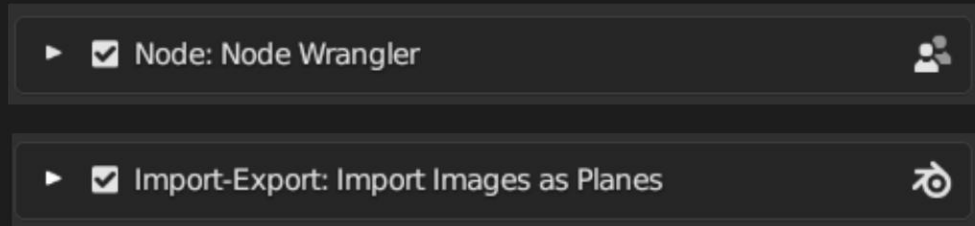
When the version after 3.5 is released, if there are fatal changes that can not follow the lecture, we will inform you later.

language

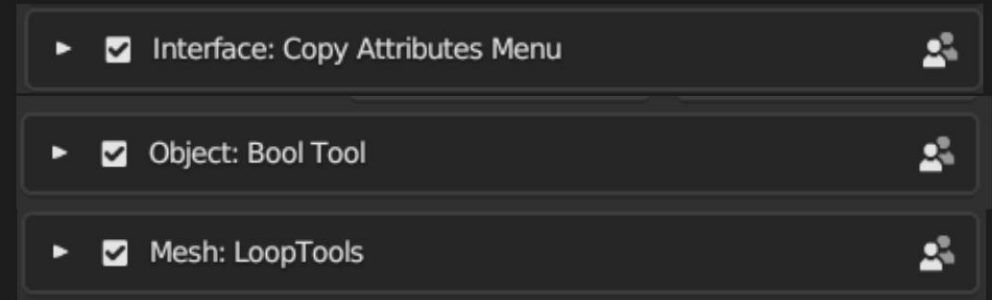


ready(2)

The following Addons are assumed to be turned on

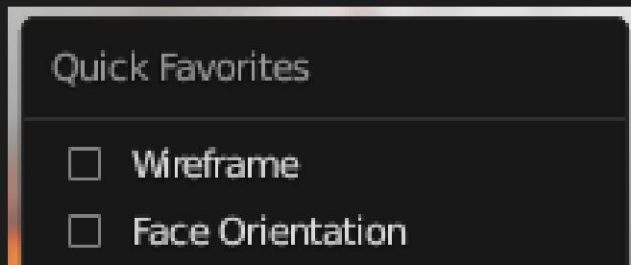


Other addons I use (not used often in this lecture)



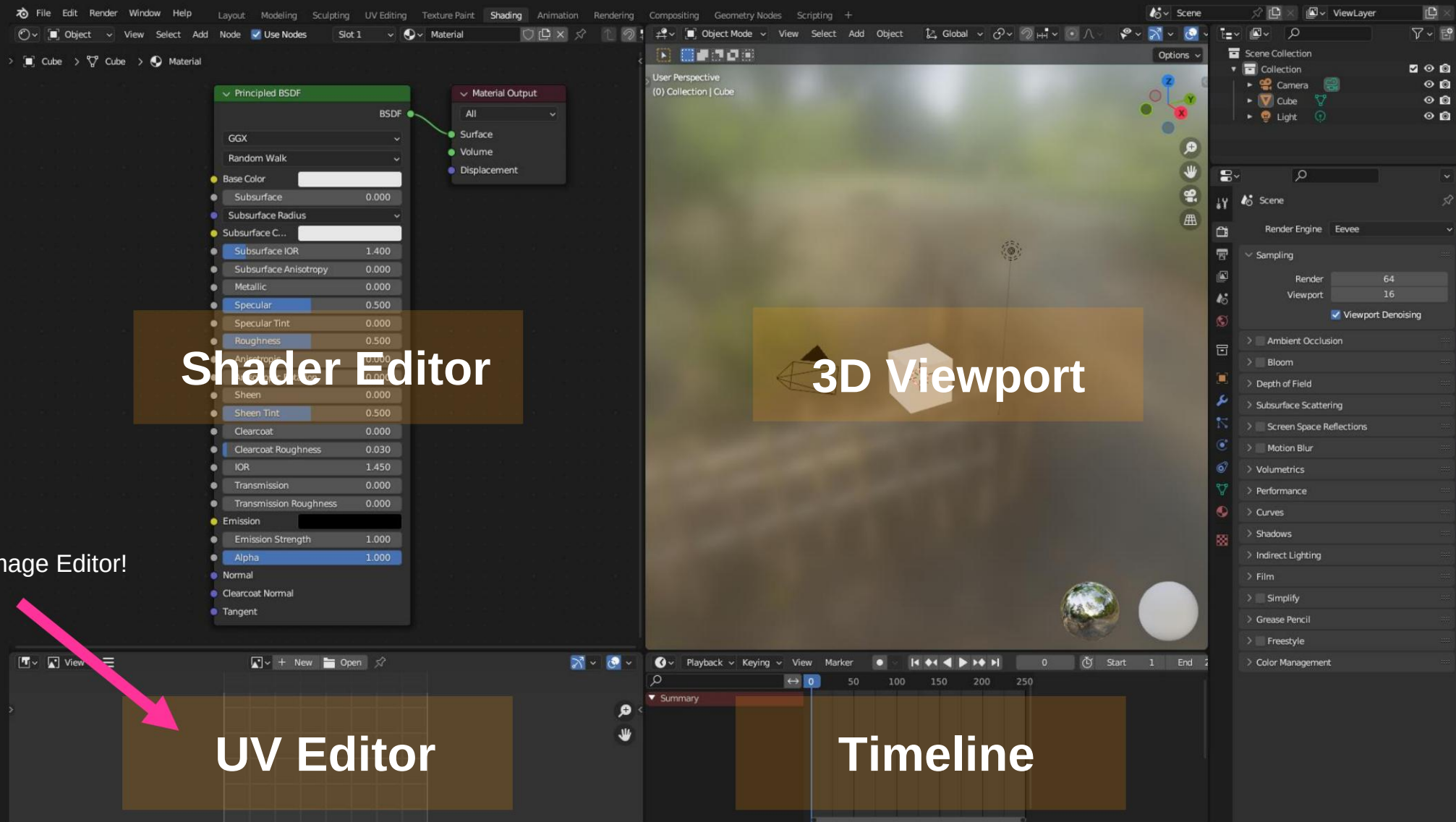
When using other add-ons, we will inform you during the lecture.

Quick Favorites settings

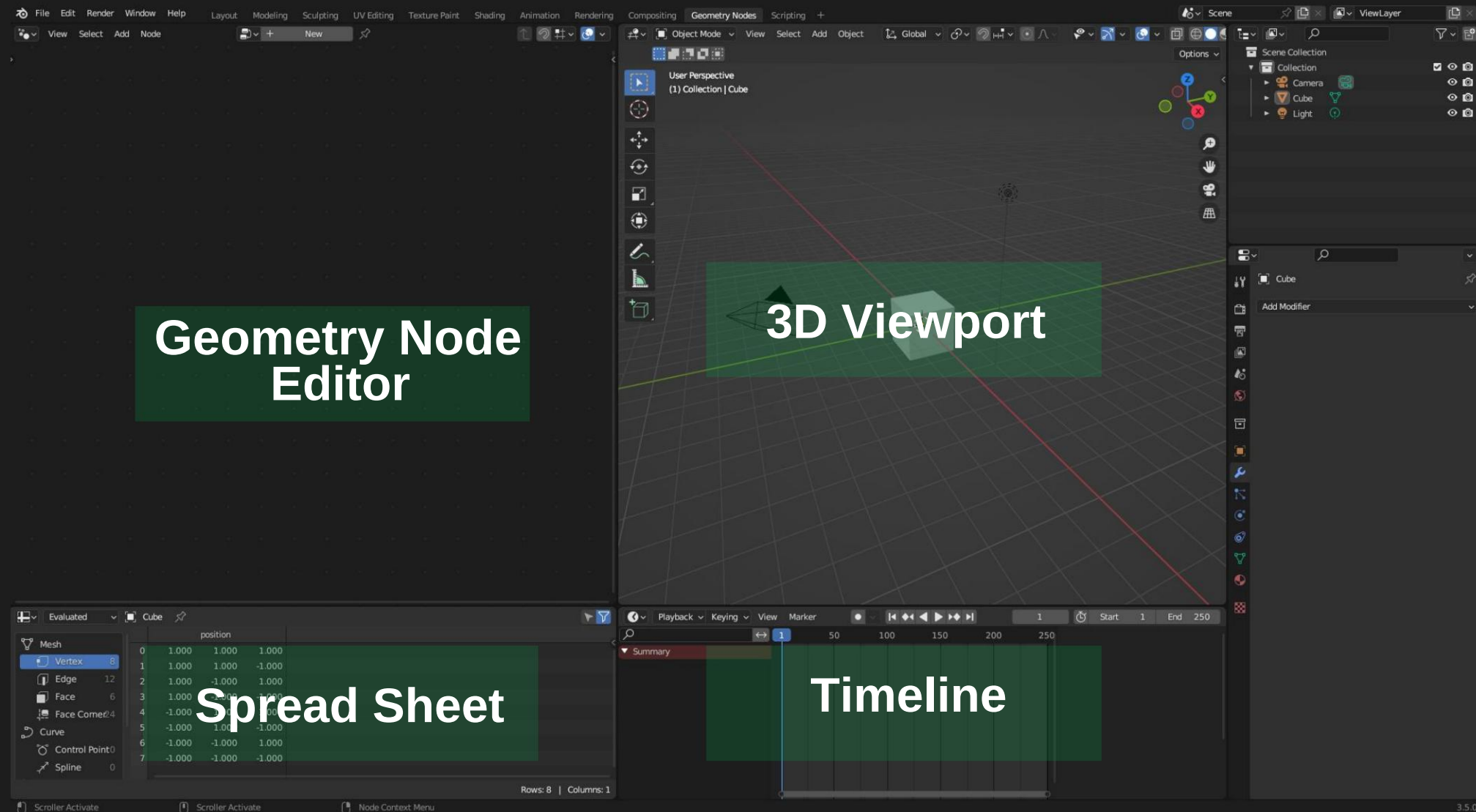


Set Wireframe and Face Orientation in both Object Mode and Edit Mode.

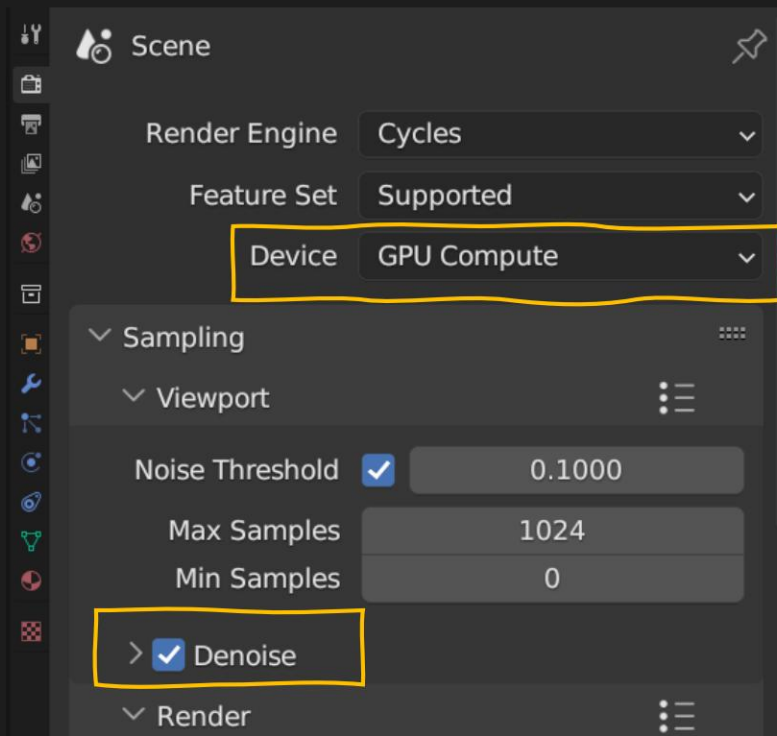
Preparation (3) Workspace setting-Shading



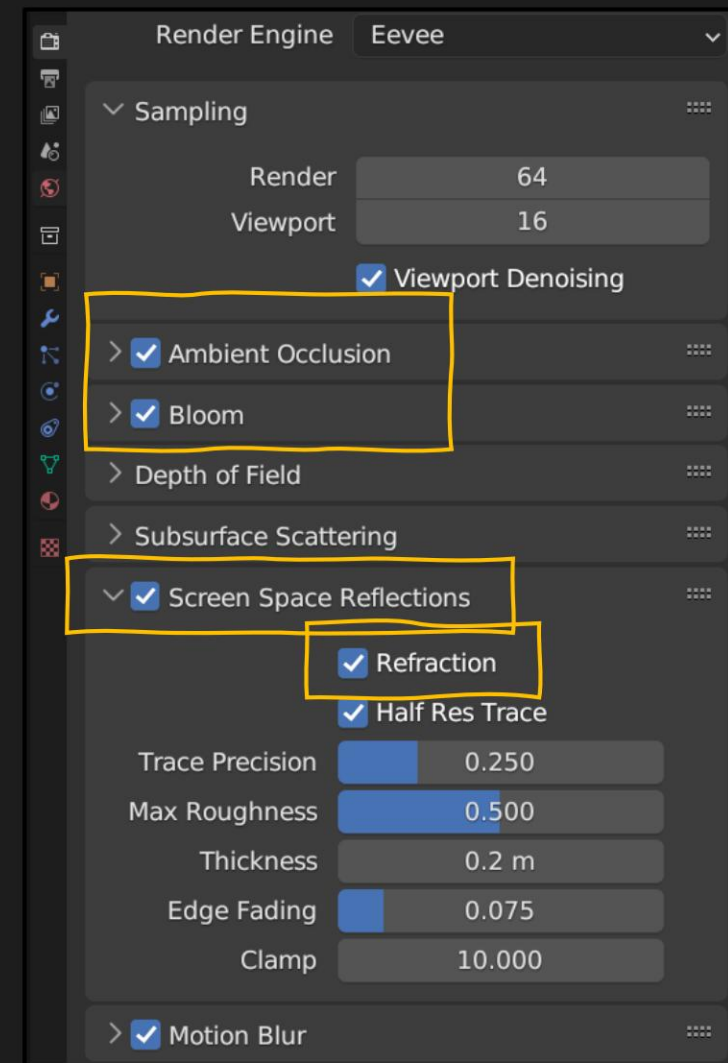
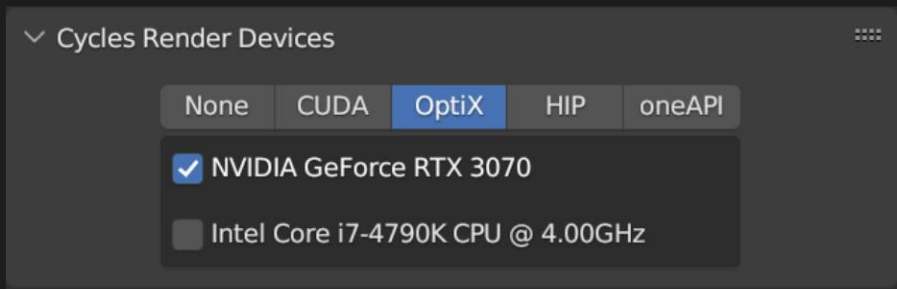
Preparation (3) Workspace Setup-Geometry Nodes



Preparation (4) Other settings

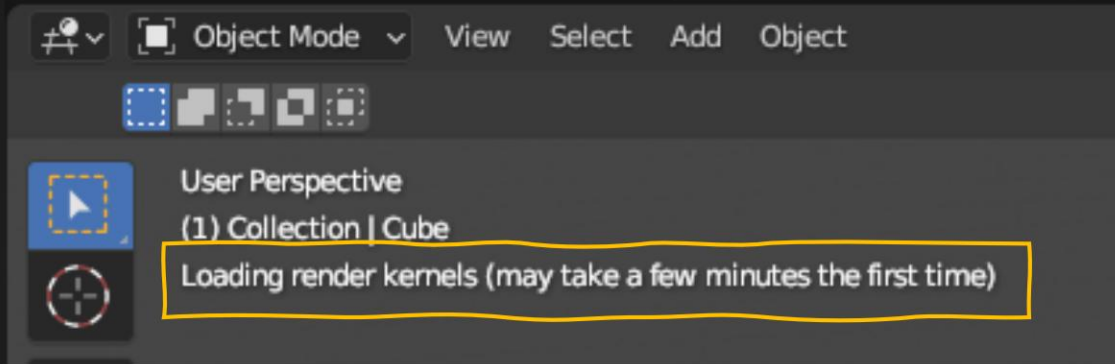


Edit-Properties-System : Please check the settings suitable for your graphics card.
Nvidia is Optix, AMD is HIP.



After change File – Defaults – Save Startup File

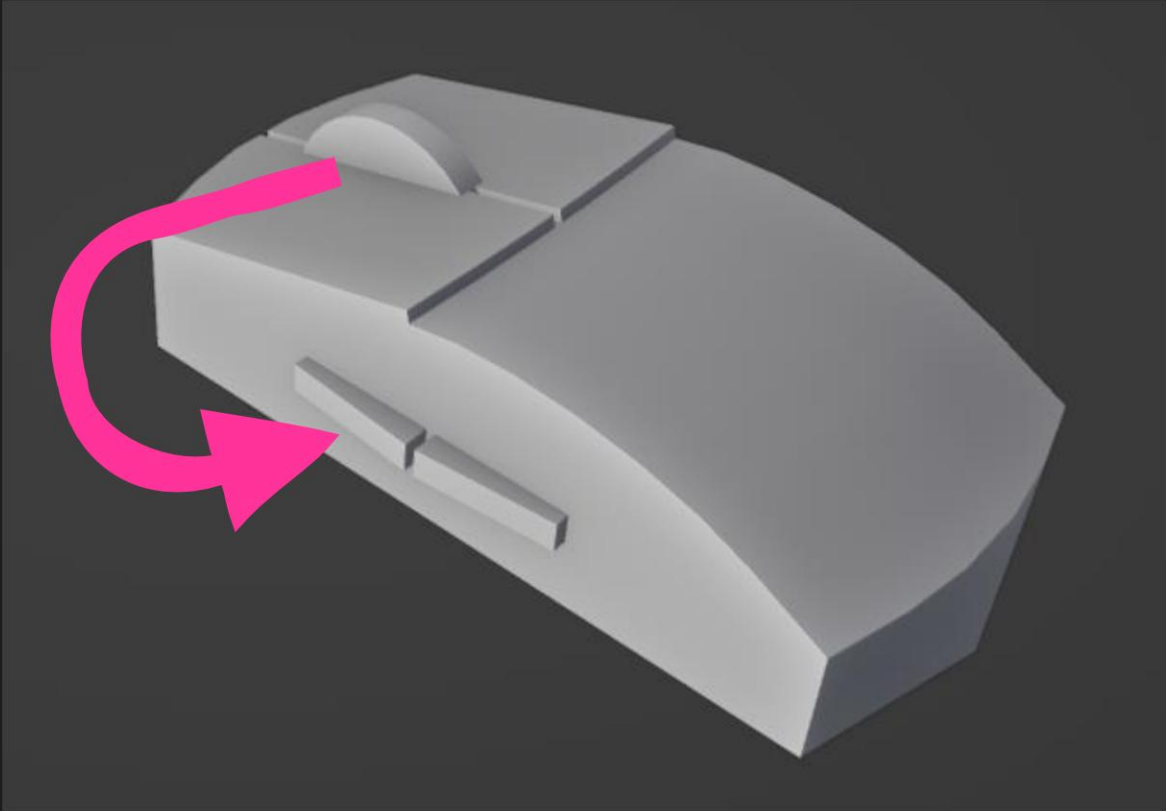
Preparation (5) Other items to check



When installing for the first time or upgrading the version, there are times when rendering stops with the message 'Loading render kernels' in Cycles render. This can take a few minutes or even tens of minutes, usually when you first connect a particular shader node.

After initial installation, this message may be displayed 2 or 3 times. If you wait patiently, it will proceed normally.

mouse settings?



Move (Rotate) Viewport The default is **the mouse wheel** button, but it can be inconvenient to press. I am using the wheel button assigned **to the mouse side button** .

Basic operation

Object mode

control keys	explanation
	Rotate the screen by dragging the mouse wheel
Shift+mouse wheel drag	screen move
Mouse wheel	screen zoom
T	Tool (left tool window)
N	Context (right setting window)
A, Alt+A	select all, deselect
Shift+A	create object
Shift+D	duplicate object
Ctrl+Z	Undo
Ctrl+Shift+Z	Undo of Undo! (= redo)
	Delete X object
G	move
R (RR)	rotation (trackball rotation)
	S-object enlargement, reduction
	Move along the X,Y,Z,X,Y,Z axes in the G,R,S state

Edit mode

control key	explanation
	Tab object - edit mode toggle
Shift+A	create mesh
	Point, line, plane selection mode in order of 1, 2, 3
X	delete menu
Ctrl+X	Dissolve Vertices
G,R,S	Move, Rotate, Scale
Shift+,ZZ in G state	to move 'except' axis
Ctrl+R	Loop Cut
	Number of Mouse Wheel Loop Cuts after Ctrl+R
E	Extrude
Alt+E	Extude menu window
I (I,I)	Inset (mode change)
Ctrl+B (Ctrl+Shift+B)	Bevel (Vertex mode)
	Shift-click multiple selection
Ctrl + Click	Select Shortest Distance

Basic operation

Camera settings

control keys	
Description	numpad 0 switch to camera view
Ctrl + Alt + numpad 0	set camera to current view

The basic operation of blender is not explained during the lecture.
If you are not familiar with the operation, it is recommended to check the table and practice at least once.

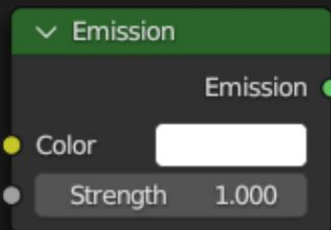
002 Basic Shader Node Operation and Basic Usage

Shader Editor, Material Structure

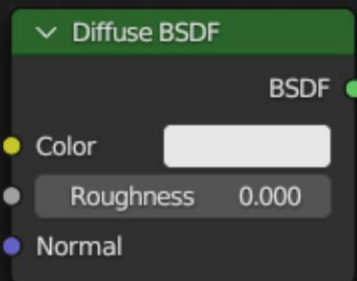
Various Shaders

Connection of Shader Nodes

Shaders Various shaders

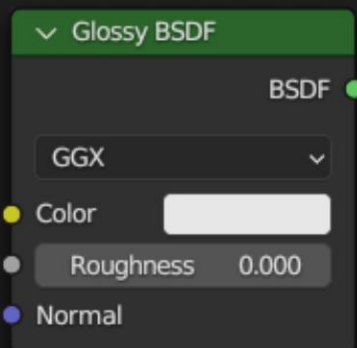


The input Color (color, image, etc.) is exported as it is with only the brightness changed. It is used for the purpose of luminous body or auxiliary lighting.

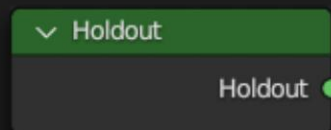


Creates a rough surface.

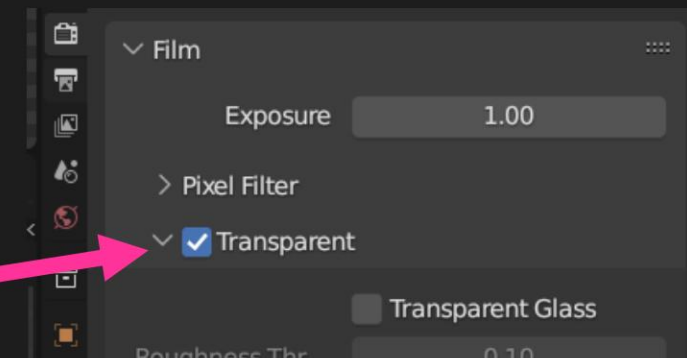
(Raising the Roughness number will make a rough surface a 'rougher' surface. This number only works for cycles.)



A shader with a slick metallic feel.



A mysterious shader where the 'render image' of the part using the material becomes transparent. Note: If you want to make the background transparent, turn on Film - Transparent in the render settings.



socket type



Float

real numbers (ex: 1, -1, 0.1, -0.31418.....)



Color

RGB color data (ex: red: (1,0,0), blue: (0,0,1)...)



Vector

XYZ vector data (ex: up direction (0,0,1), right direction (1,0,0)...)

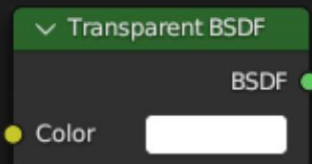


Shader

shader data.

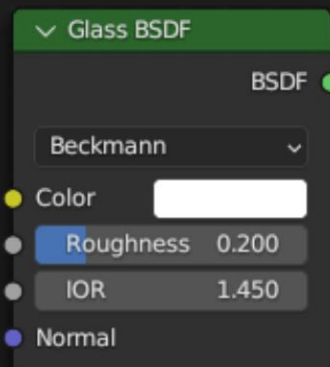
Shaders

Some shaders require setup when used in Eevee



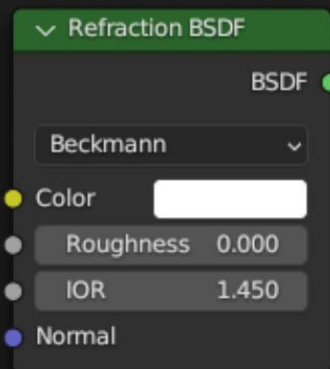
A shader representing transparency.

In Eevee, you need to set Blend mode in Options to Alpha xxx to work.

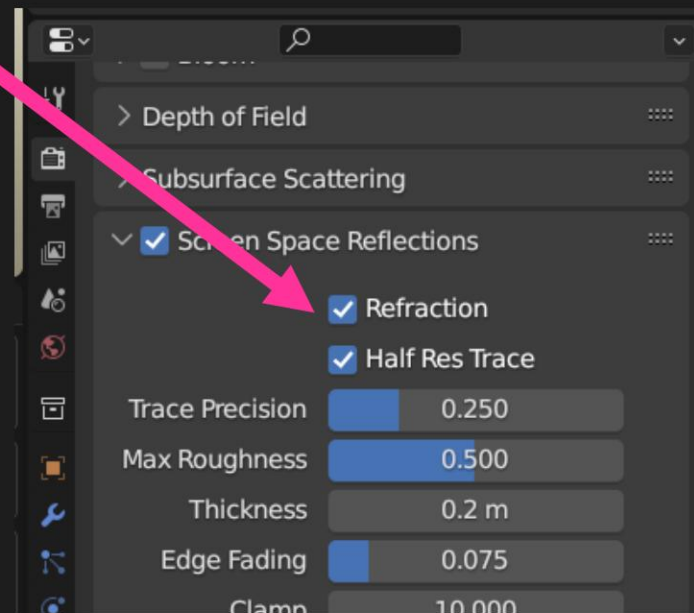
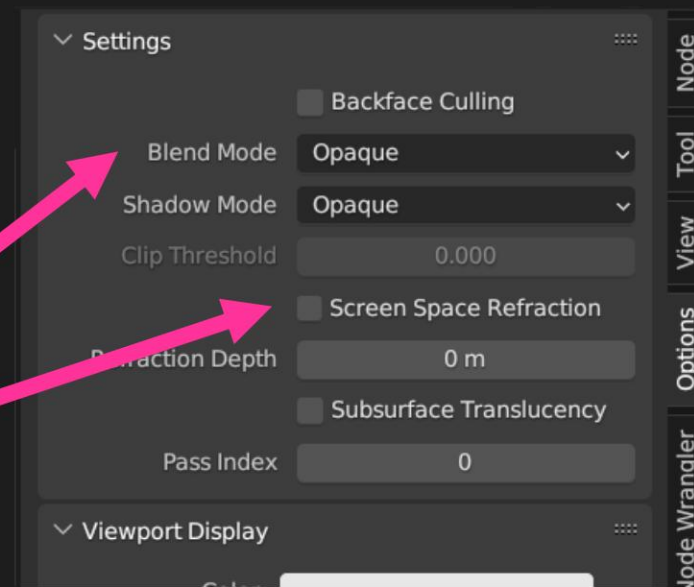


A shader representing glass.

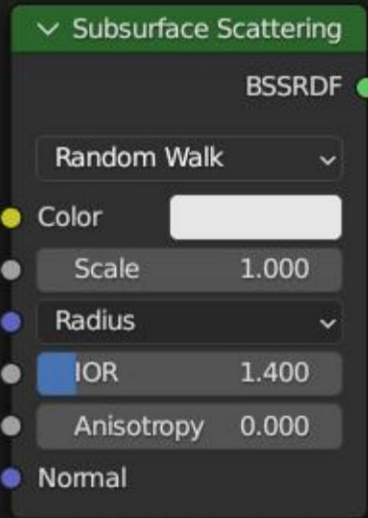
Eevee requires Screen Space Refraction to be turned on for this to work.



Similar to Glass BSDF, but expresses only 'refraction'.
There is no reflection from the surface.



Shaders Various shaders

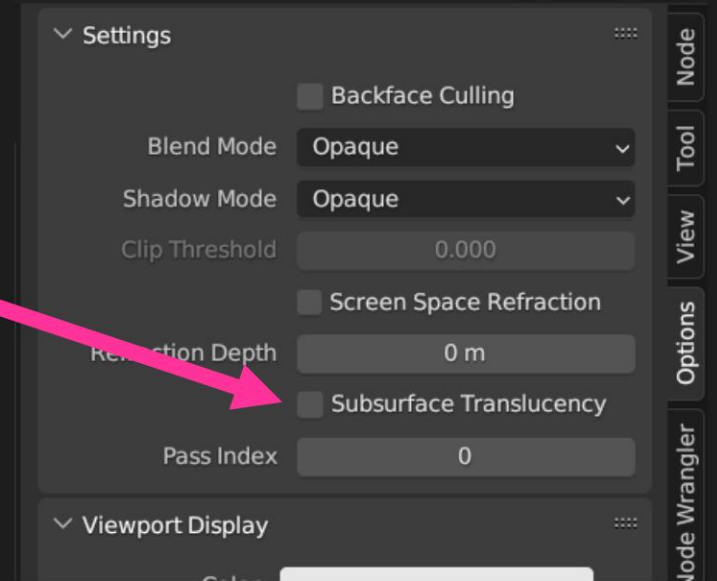


Shaders used to create subtly transparent materials such as skin. Eevee requires Subsurface Translucency to be turned on for this to work.

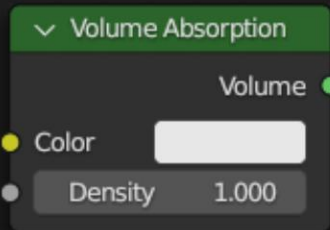
Anisotropy specifies the directionality of transmission. It doesn't have much of an impact. (Refer to the Volume lecture later for directionality.)



A shader that expresses the penetration that occurs on a rough surface. Note that there is no option to adjust penetration depth or roughness.

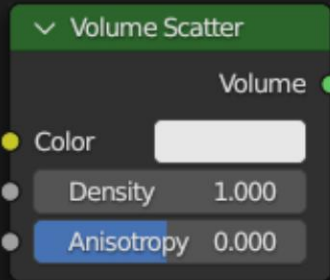


Shaders Various shaders



Creates a volumetric effect that darkens with distance and depth.

Plug the Volume node into the Volume socket of Material Output.

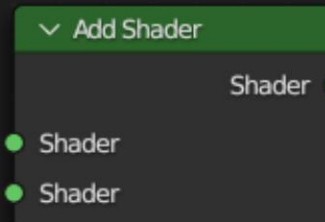


Light scatters as it passes, creating a visible effect. The effect is maximized by the angle and position of the external lighting.



Composite the two input shaders!

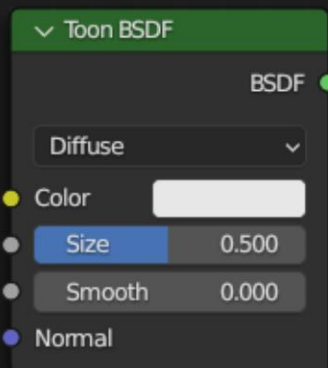
Smaller Factor values blend more of the upper shader and larger values of the lower shader.



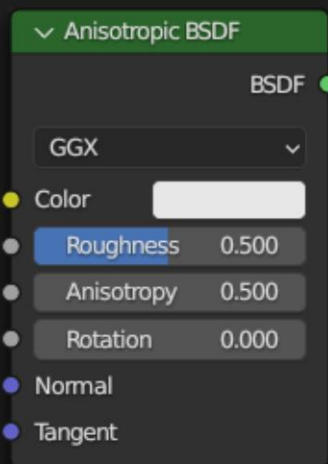
Adds the two input shaders. Similar to Photoshop's add blending mode. More detailed explanation in the video texture lecture.

Shaders

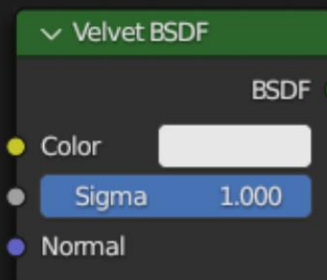
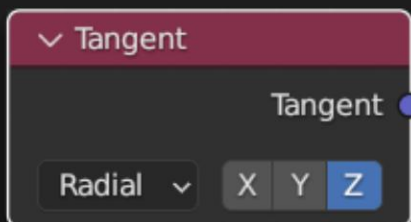
There are also shaders that only work in Cycles



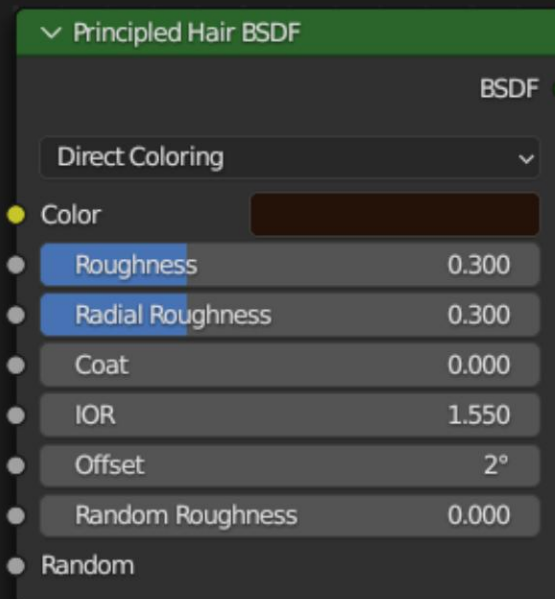
A shader that creates a cartoon-like expression. Effects like this are less common as Eevee handles them better.



Expresses a surface with roughness in a specific direction, such as brushed metal. If you want to change the direction, you need to input an additional direction into the Tangent socket.



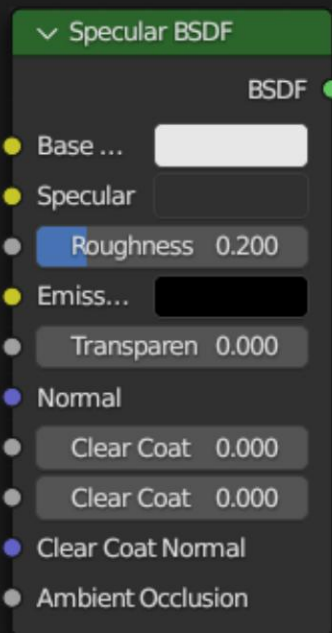
It expresses the edge of the cloth material. It's technically outdated, so it's not used very well.



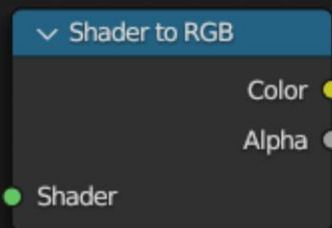
A hair material to apply to Hair Particles (or Hair Curves in newer versions). For details, see the Hair lecture in Geometry Nodes.

Shaders

There are also shaders that only work on Eevee



Surface color and reflected color can be specified separately.



It's not a shader, it's a unique node that transforms a shader into an image. (refer to the NPR lecture)

Principled Shader A

shader that incorporates (almost) everything.

Principled BSDF

BSDF

Texture

Random Walk

Base Color

Subsurface0.000

Subsurface Radius

Subsurface C...

Subsurface IOR1.400

Subsurface Anisotropy0.000

Metallic0.000

Specular0.500

Specular Tint0.000

Roughness0.500

Anisotropic0.000

Anisotropic Rotation0.000

Sheen0.000

Sheen Tint0.500

Clearcoat0.000

Clearcoat Roughness0.030

IOR1.450

Transmission0.000

Transmission Roughness0.000

Emission

Emission Strength1.000

Alpha1.000

Normal

Clearcoat Normal

Tangent

Principled Volume

Volume

Color

Color Attribute

Density1.000

Density Attrib...density

Anisotropy0.000

Absorption C...

Emission Strength0.000

Emission Color

Blackbody Intensity0.000

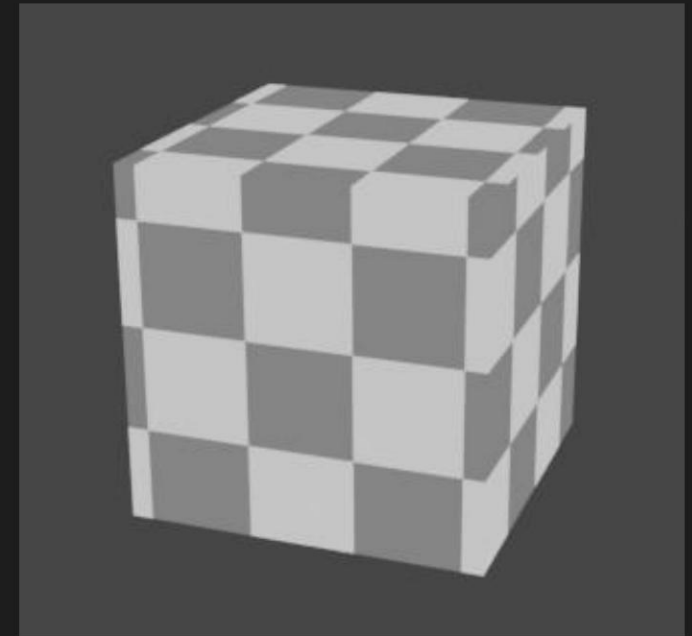
Blackbody Tint

Temperature1000.000

Temperature ...temperature

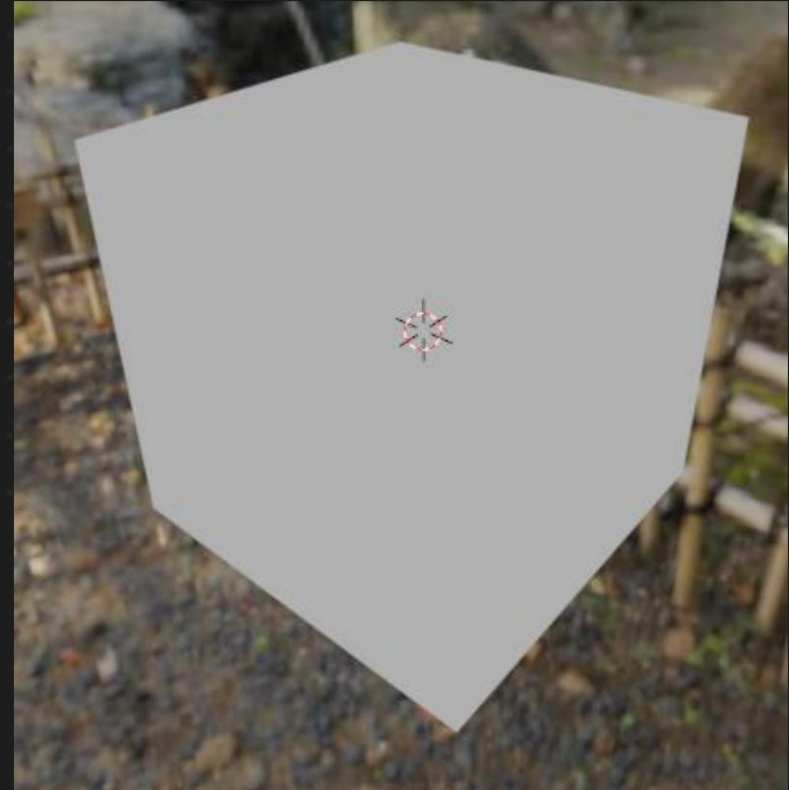
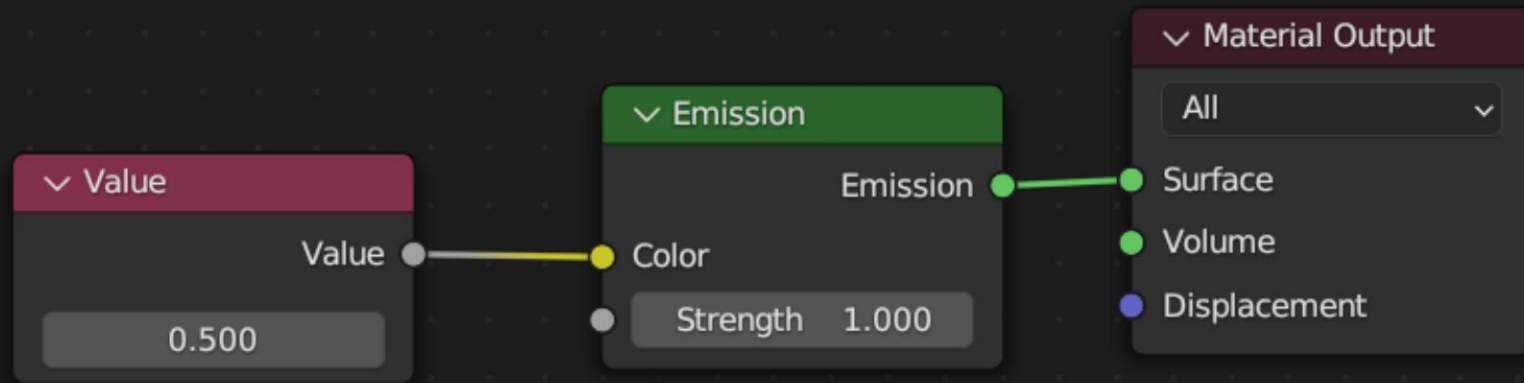
Default connection:

Texture - Shader - Material Output



If you plug Float **into Color** ?

If a float value is inserted into a color, the same value is input to each RGB, and achromatic color is output.

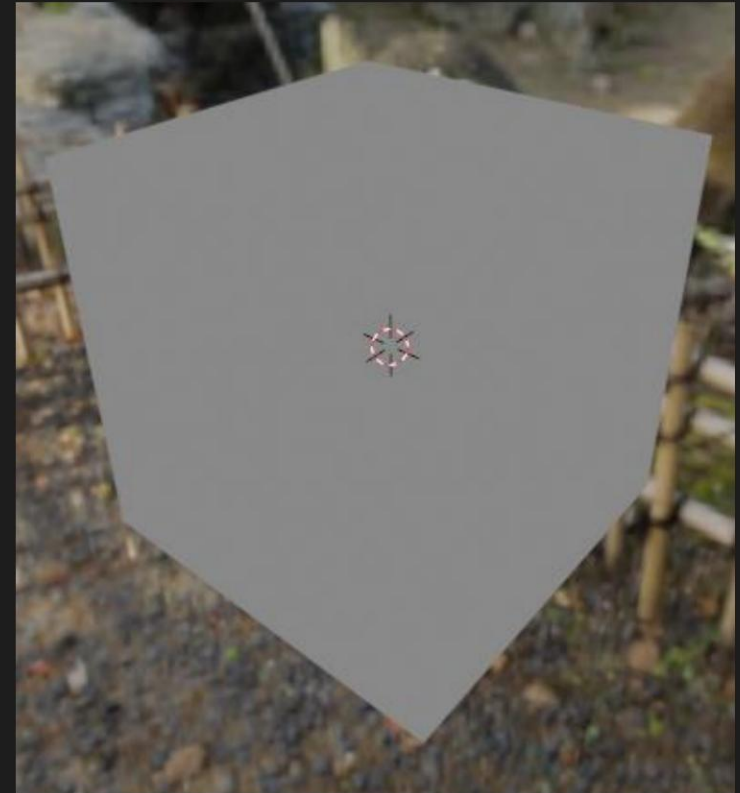


If you plug **Color** into Float?

If you plug Color into Float, the 'brightness' of Color is output.



(1,0,0) → 'brightness' of red = 0.2126..



socket preview

Limited to Material Output, any data type can be plugged in.
It is the same effect as plugged into Emission's Color.



003 Vector

Texture

Coordinates - How images are applied to a 3D surface

- Meaning and characteristics of UV, Generated, and Object coordinates
- Mapping method of Image Texture

Normal

- Normal map

How to superimpose a 2D image into 3D?

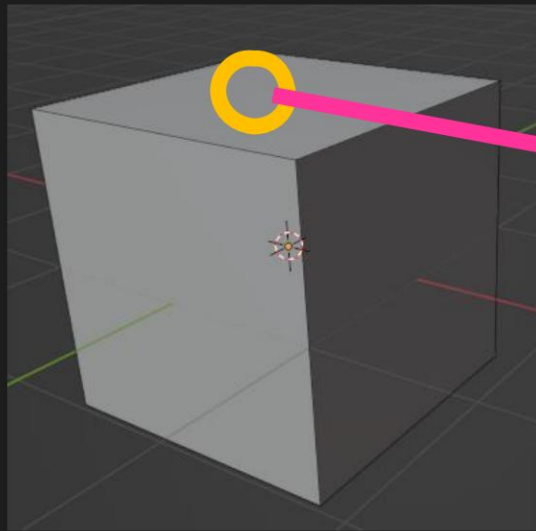
UV

A principle similar to development

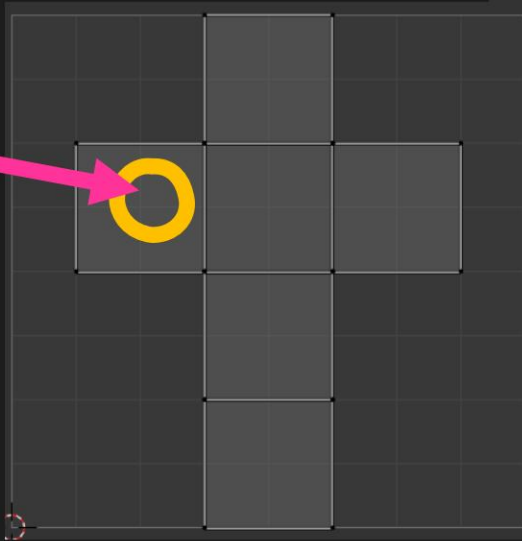


How is the 2D development map connected to the 3D?

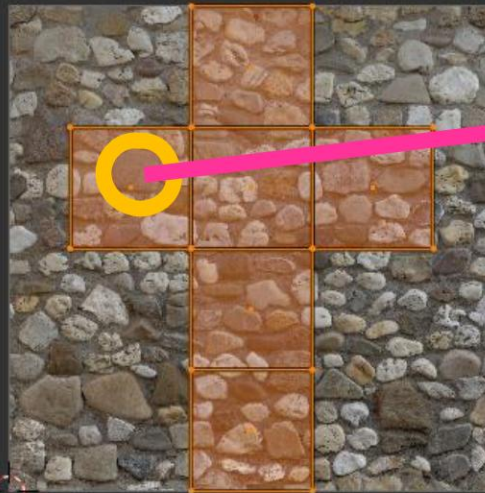
The location created by the UV map corresponds to the 3D location.



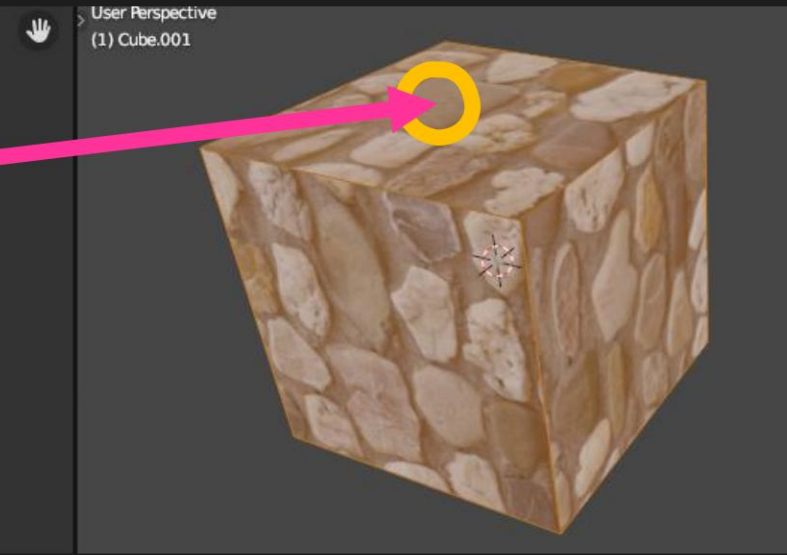
this point...



Corresponds to a point on
a two-dimensional plane.



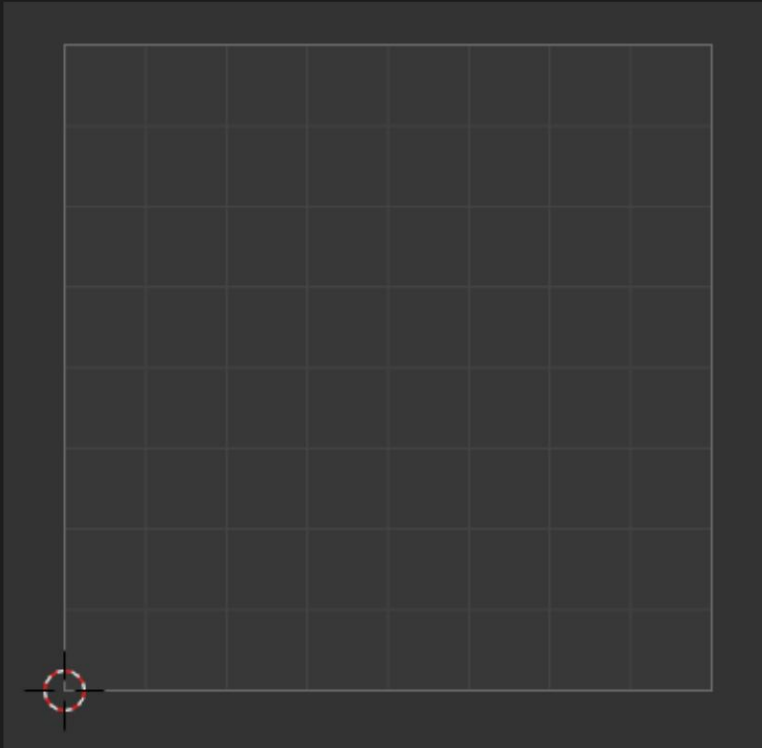
Read the color of
the image at that point.



that color on the
surface .

How is the 2D development map connected to the 3D?

1. UV coordinates



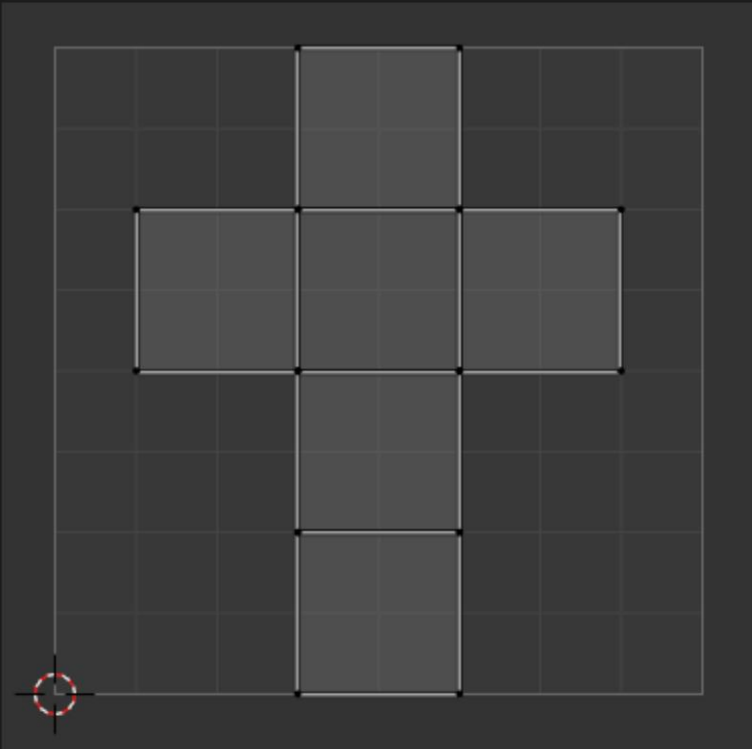
UV coordinates are two-dimensional coordinates used by images. Since

this is different from the three-dimensional space created by the XYZ axes, X, Y, and Z are not used, and new alphabets, U and V, are used.

The horizontal axis is U and the vertical axis is V, and has a value between 0 and 1 in the horizontal and vertical directions.

How is the 2D development map connected to the 3D?

2. UV unwrap

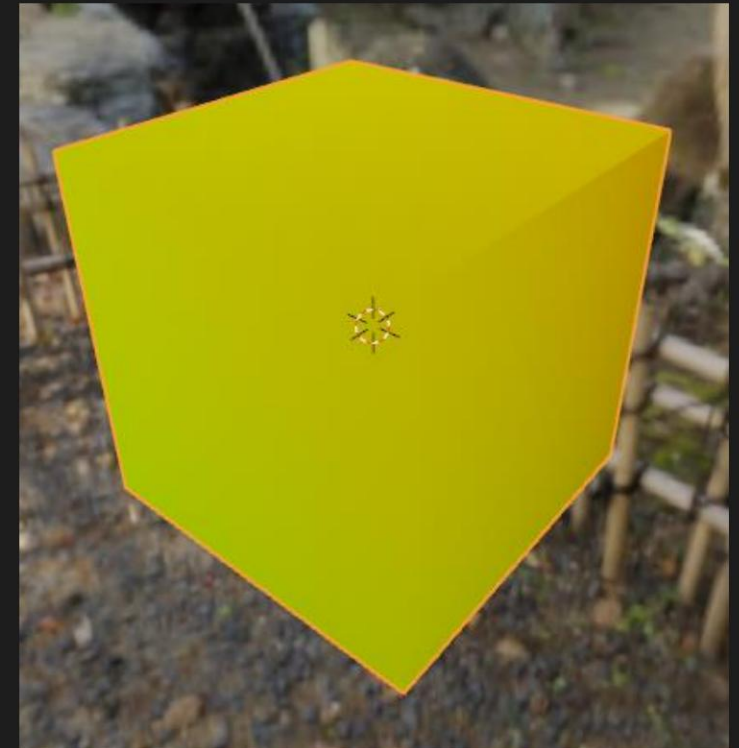
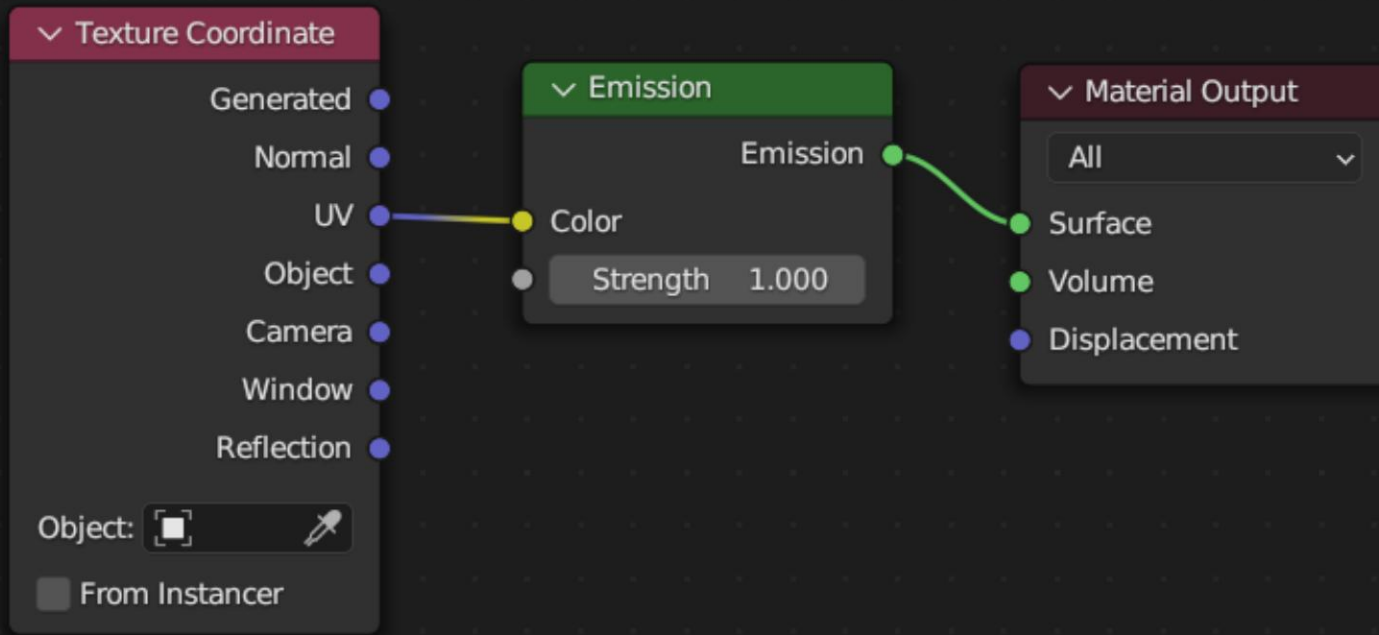


Determines where to place each face of a 3D object in the UV.

Preview of Vector

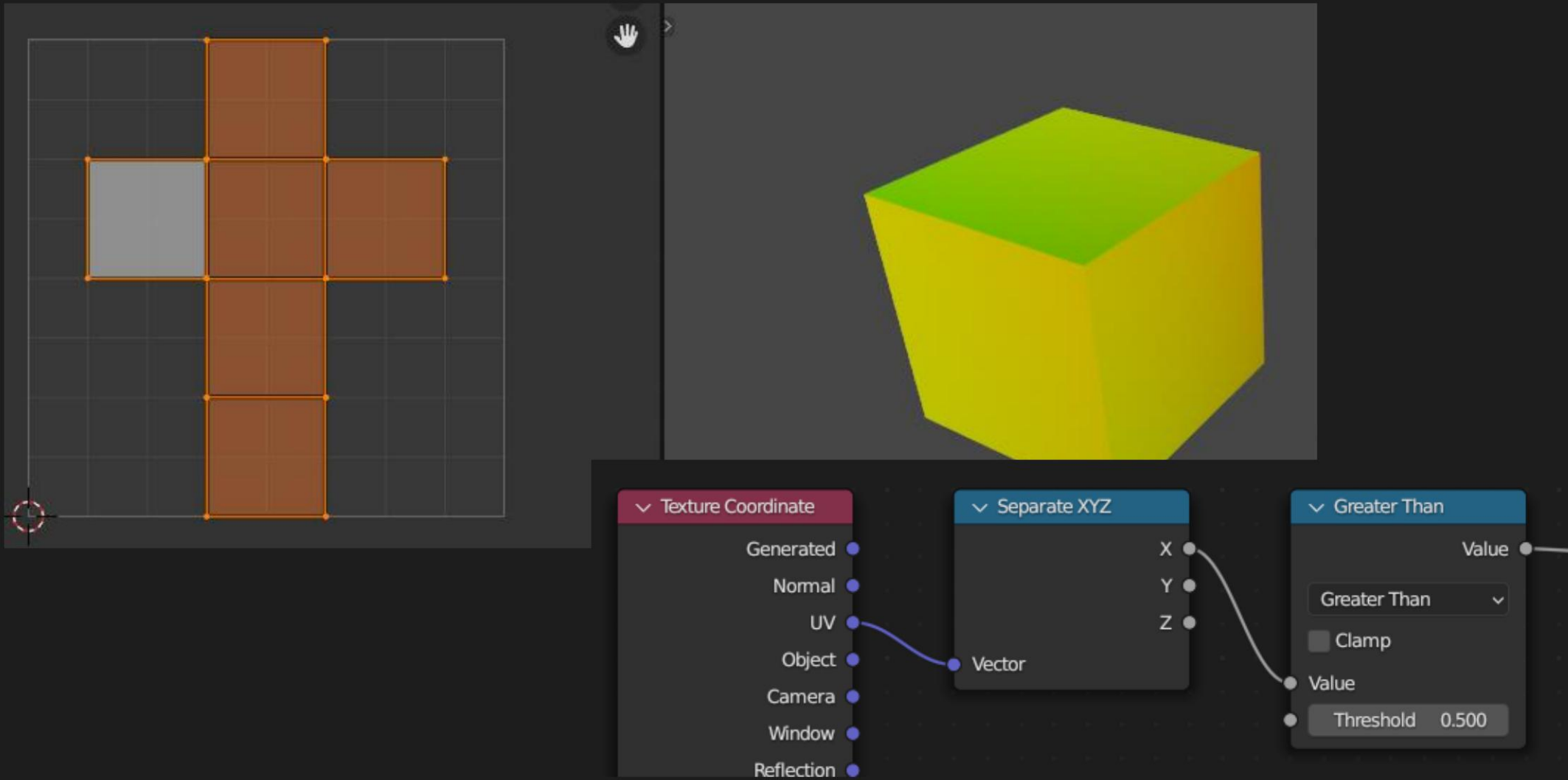
If vector values are plugged into the outputs, X, Y, Z correspond to R, G, B respectively.

That is, the X-axis represents red, the Y-axis represents green, and the Z-axis represents blue.



Color of UV coordinates (?)

UV ranges from red to green, mostly yellow. (why?) Let's try the Separate XYZ and Math nodes.

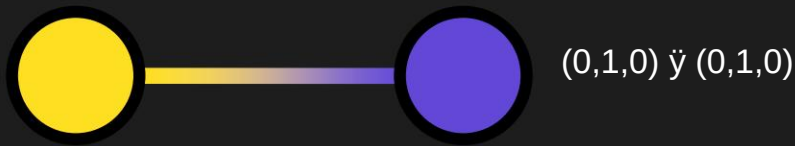


socket compatibility cleanup

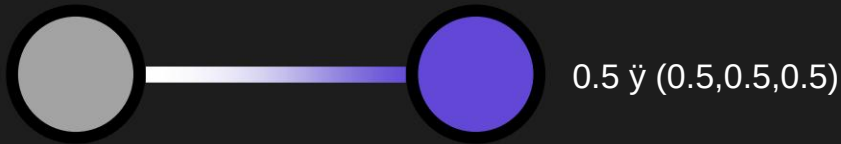
You don't have to memorize it. Just remember to be compatible in a reasonable way.



$(1,0,0) \rightarrow (1,0,0)$



$(0,1,0) \rightarrow (0,1,0)$



$0.5 \rightarrow (0.5,0.5,0.5)$



$0.5 \rightarrow (0.5,0.5,0.5)$



$(1,0,0) \rightarrow \text{'average' of } 1,0,0 = 0.333..$



$(1,0,0) \rightarrow \text{'brightness' of red} = 0.2126..$

How is the 2D development map connected to the 3D?

3. Image Texture



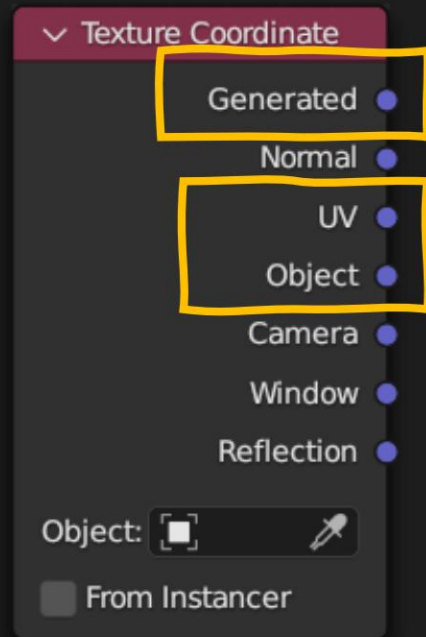
Images are placed at UV coordinates.

At this time, regardless of the image resolution, the width and height are aligned between 0 and 1.

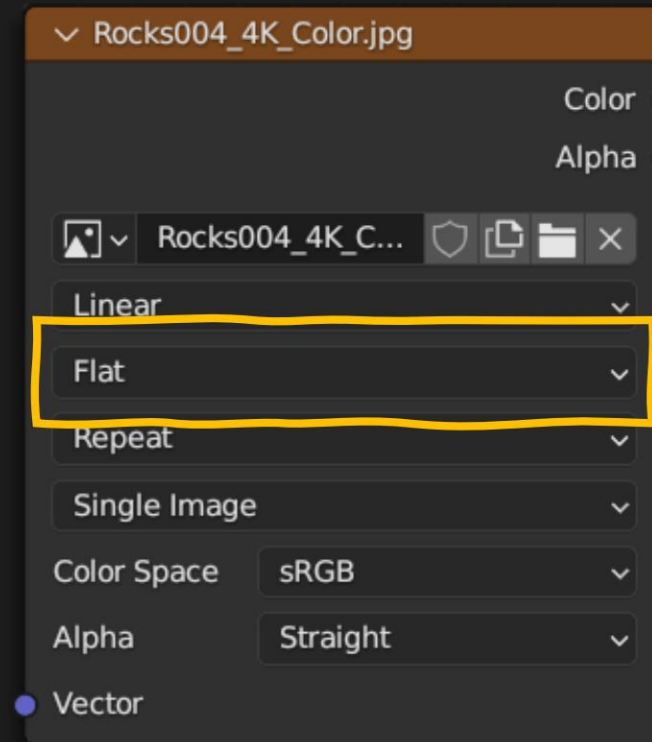
Therefore, images with different horizontal and vertical sizes will be out of proportion on the coordinates.

Coordinates and Mapping Methods

Texture Coordinates: Generated, UV, Object



Mapping method : Flat, Box, Sphere, Tube

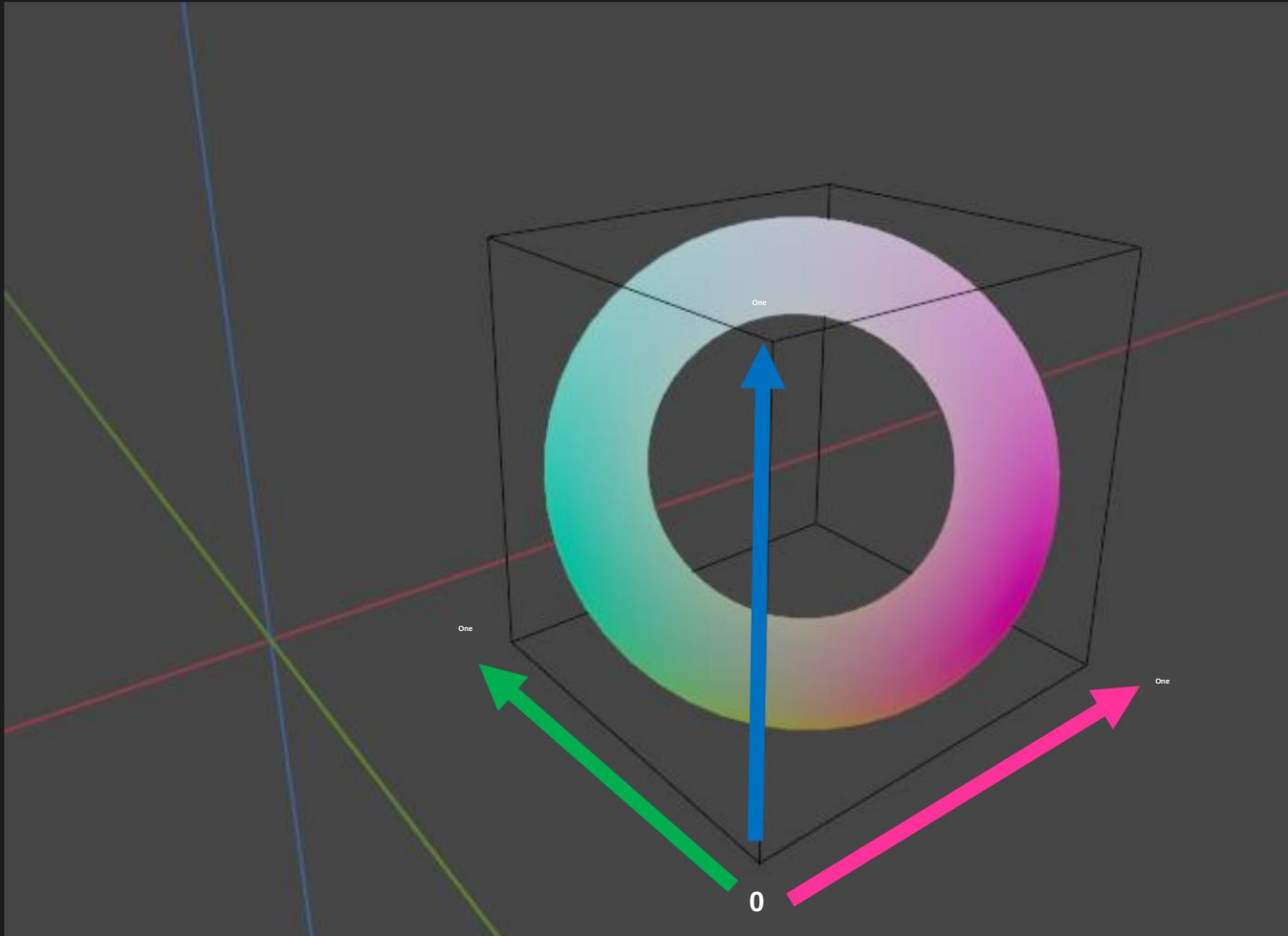


• Use the coordinates of Camera and Window directly

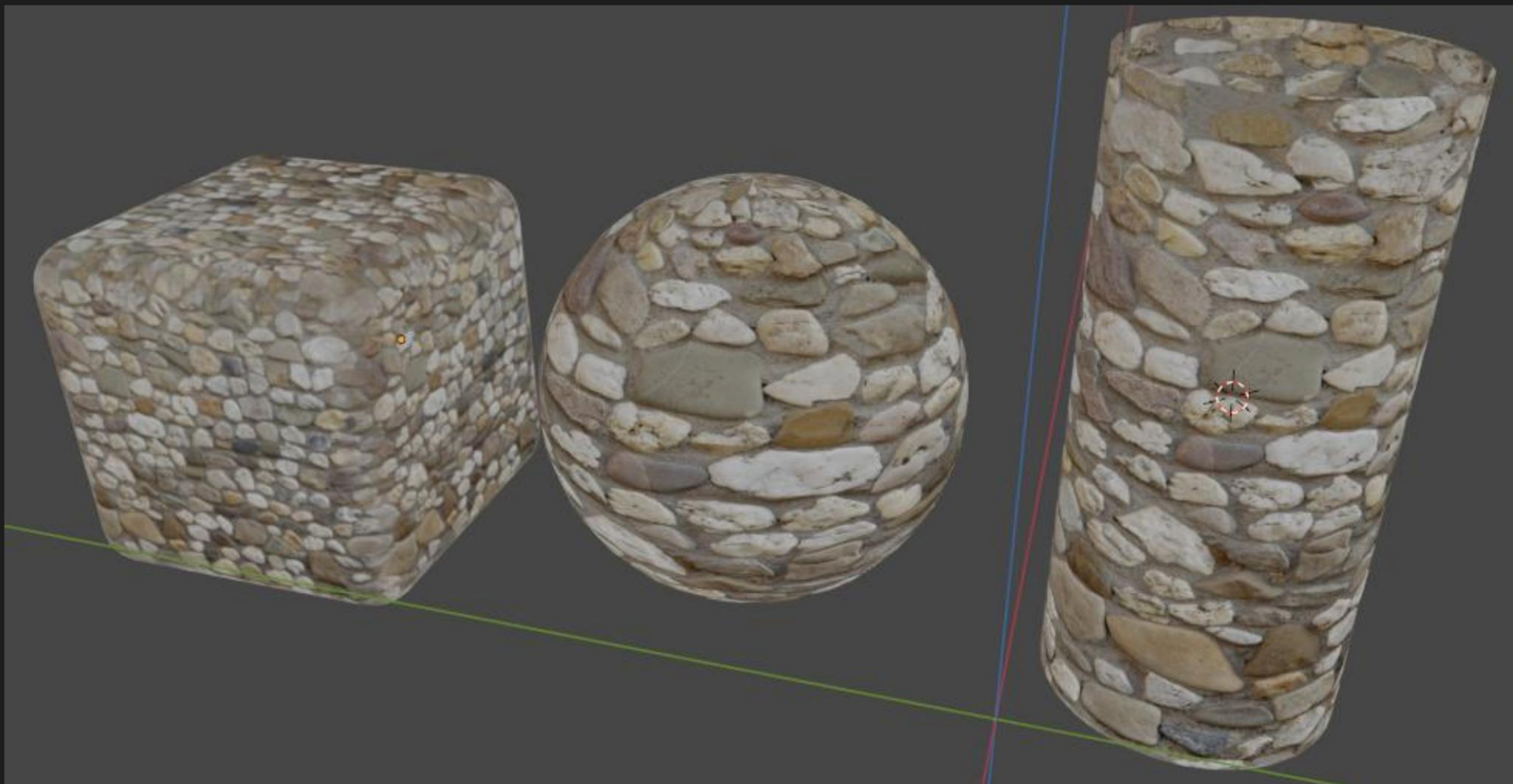
• Reflection works only when using the Environment texture. (refer to the NPR lecture)

3D generated coordinates surrounding the object

Set the width, length, and height of the Bounding Box to a value between 0 and 1



Box, Sphere, and Tube mapping for generated coordinates



Object coordinates extending from the origin

Centered on the object's origin, it extends in the X, Y, and Z directions.



Generated vs Object

	Generated	Object
center position	bounding box edge	Origin
scale	Dents depending on the size of the object	Follow object scale
modifier	Maintain original coordinates before transformation (Excluding Geometry nodes)	When the object is deformed, the texture will move. can

Normal

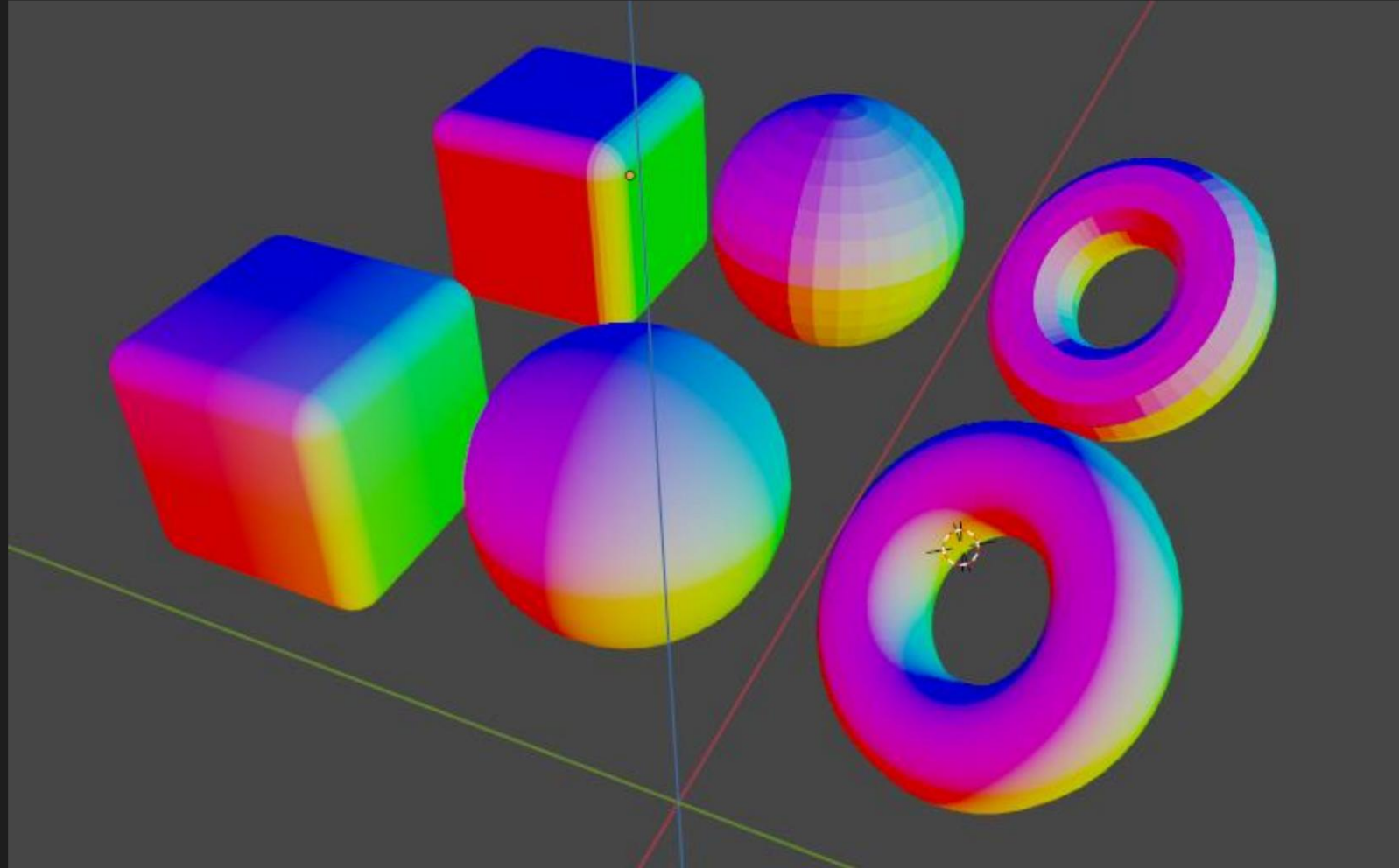
The 'direction' of the object's surface pointing outward

- Always size 1.

- Shade smooth vs Shade flat

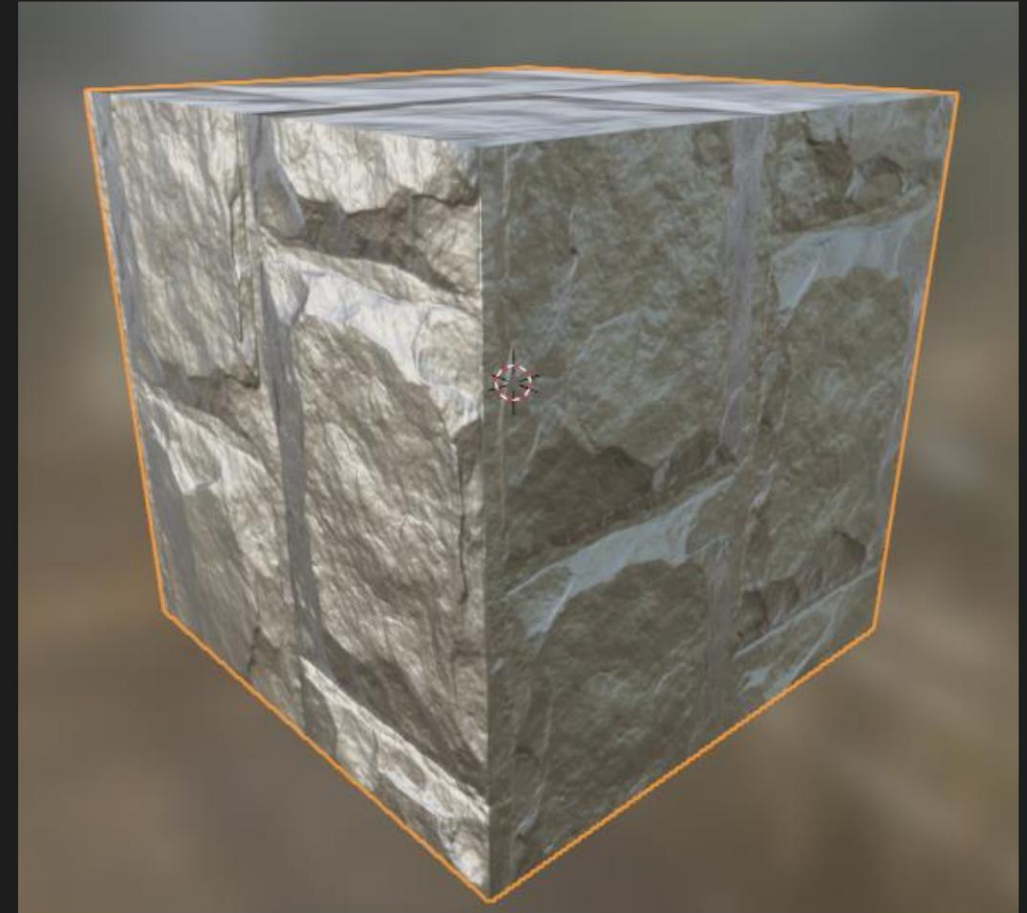
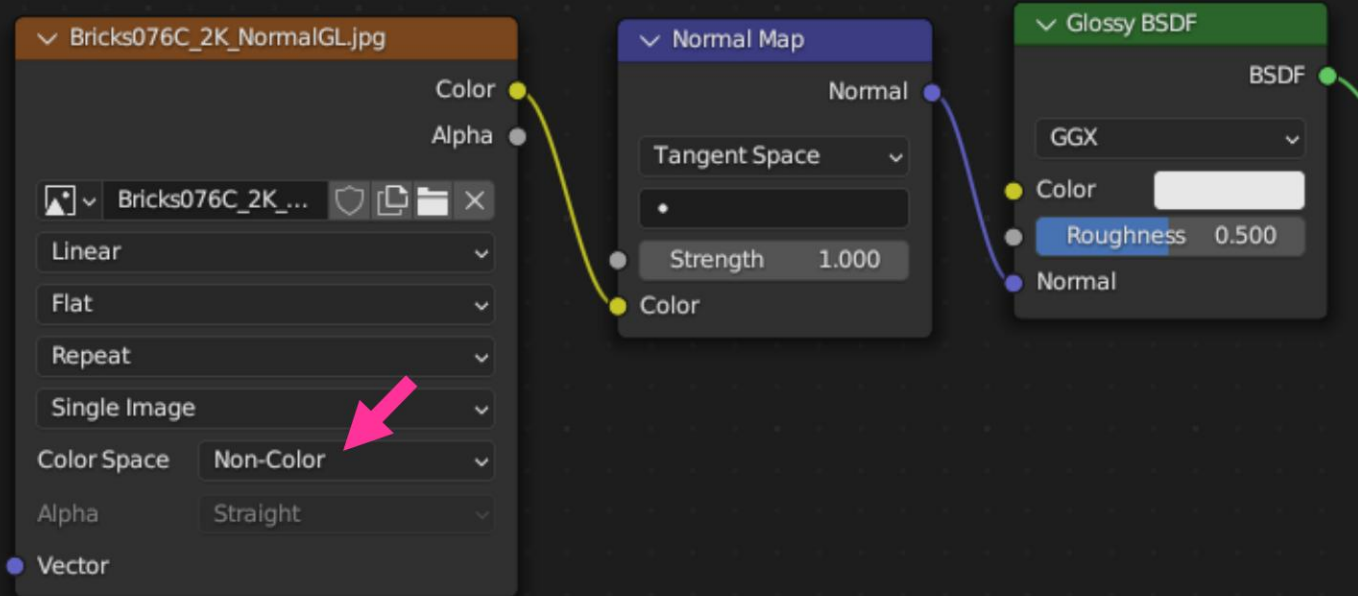
- Weighted Normal

- It doesn't have to match the direction of the real face, that's one of the important points.



Normal Map

-Can be transformed into Bump node and Normal Map.



004 Control of Mapping Nodes and Coordinates

Node animation using **texture coordinate**

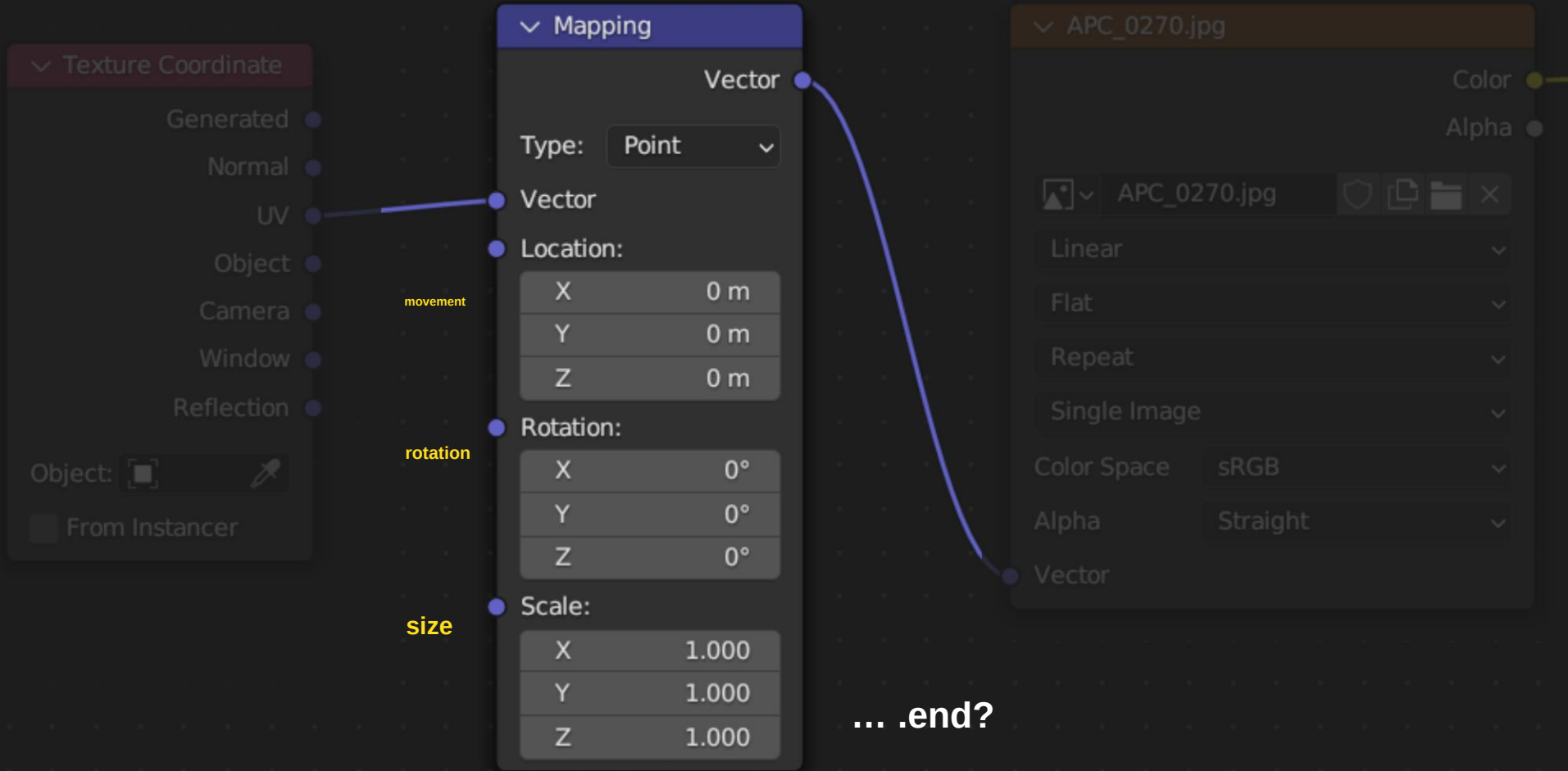
movement, rotation, and scale coordinate movement

Introducing the Mix Node



Mapping node

move coordinates



Mapping node

Texture Coordinate

Generated

Normal

UV

Object

Camera

Window

Reflection

Object:  

☐ From Instancer

PavingStones121_2K_Color.jpg

 PavingStones12...    

Linear

Flat

Repeat

Single Image

Color Space: sRGB

Alpha: Straight

Color

Alpha

Vector

Mapping

Vector

Type: Point

Vector

Location:

X0 m

Y0.1 m

Z0 m

Rotation:

X35.6°

Y-29.2°

Z0°

Scale:

X1.000

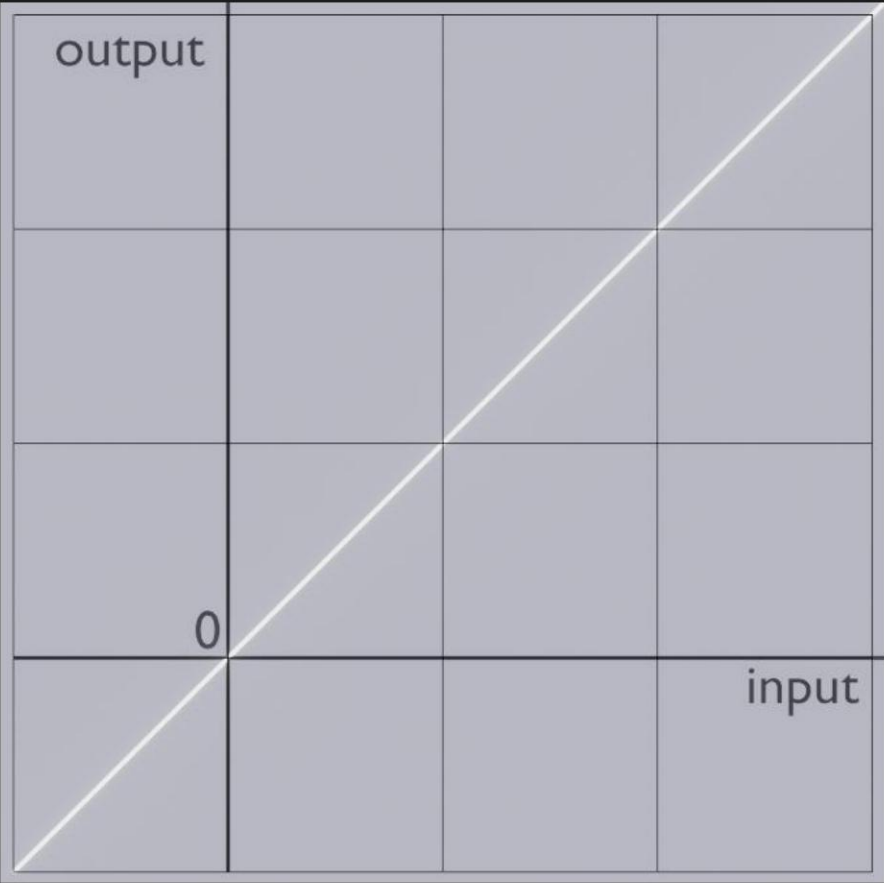
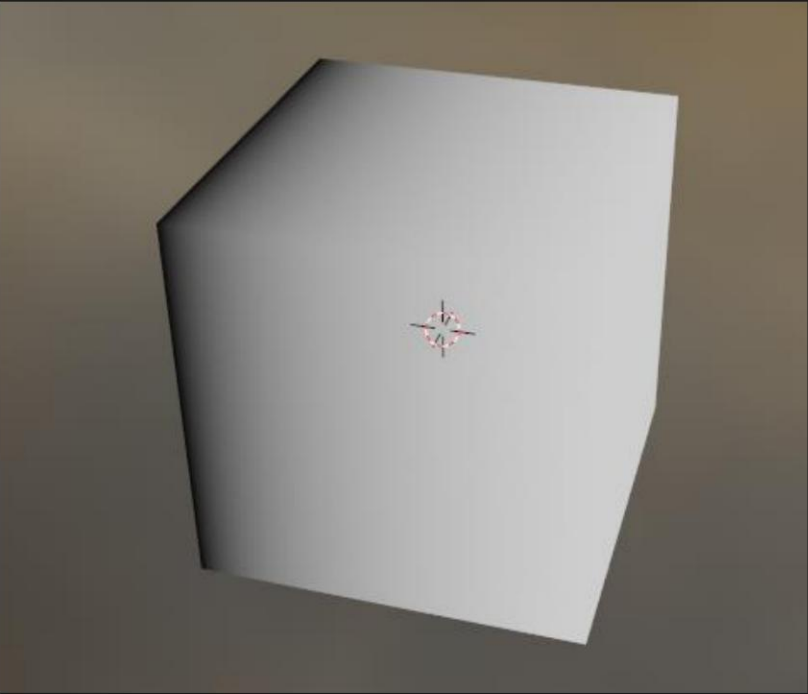
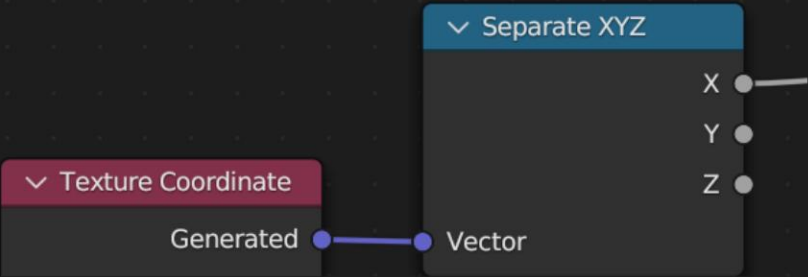
Y1.000

Z1.000



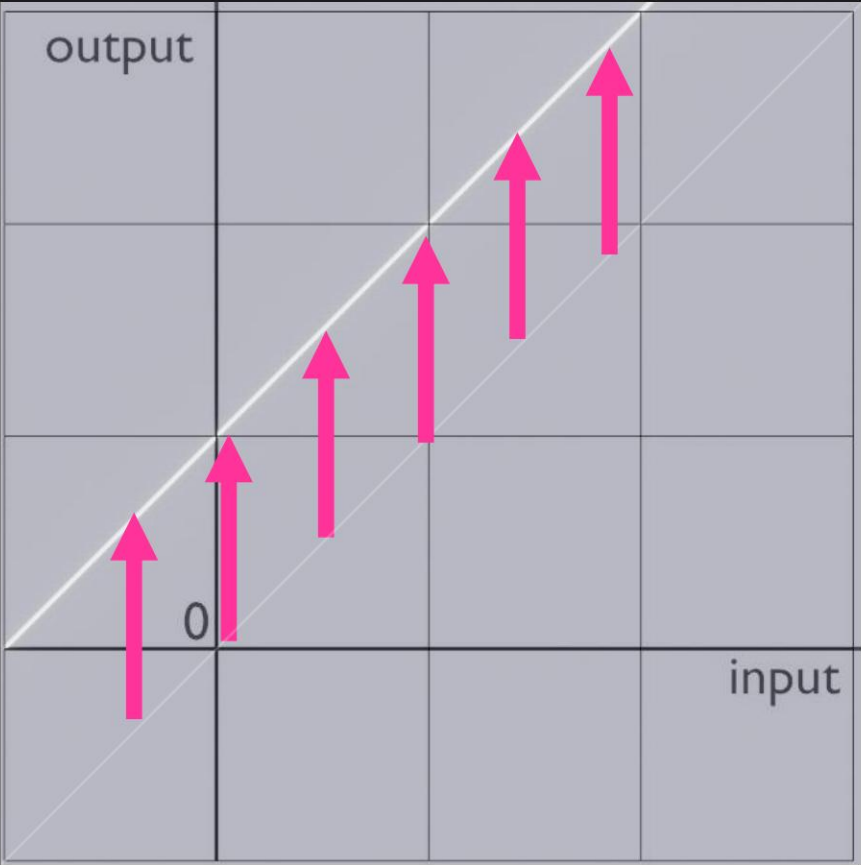
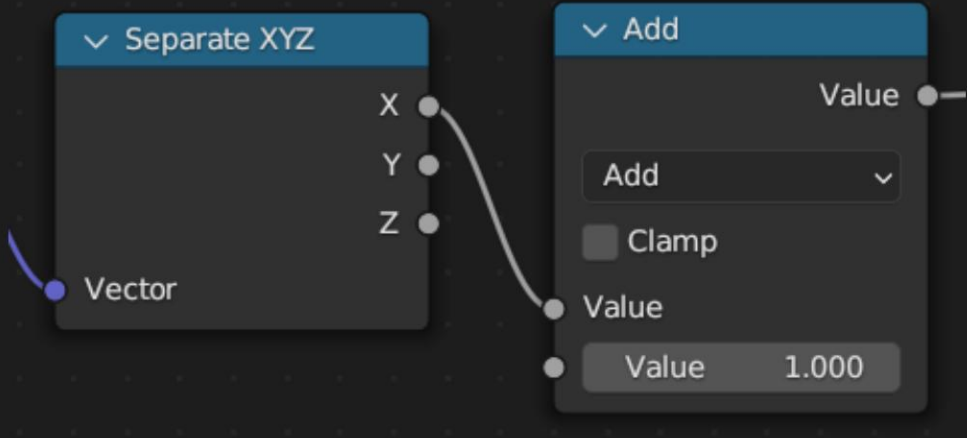
???

principle of movement



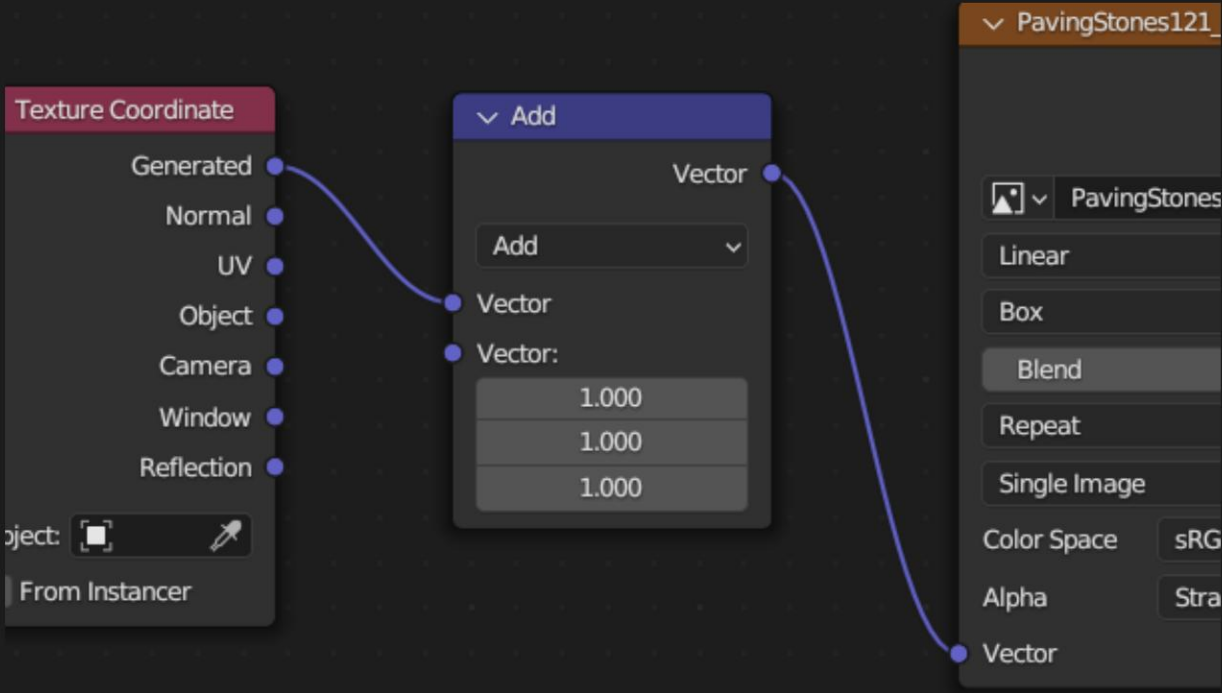
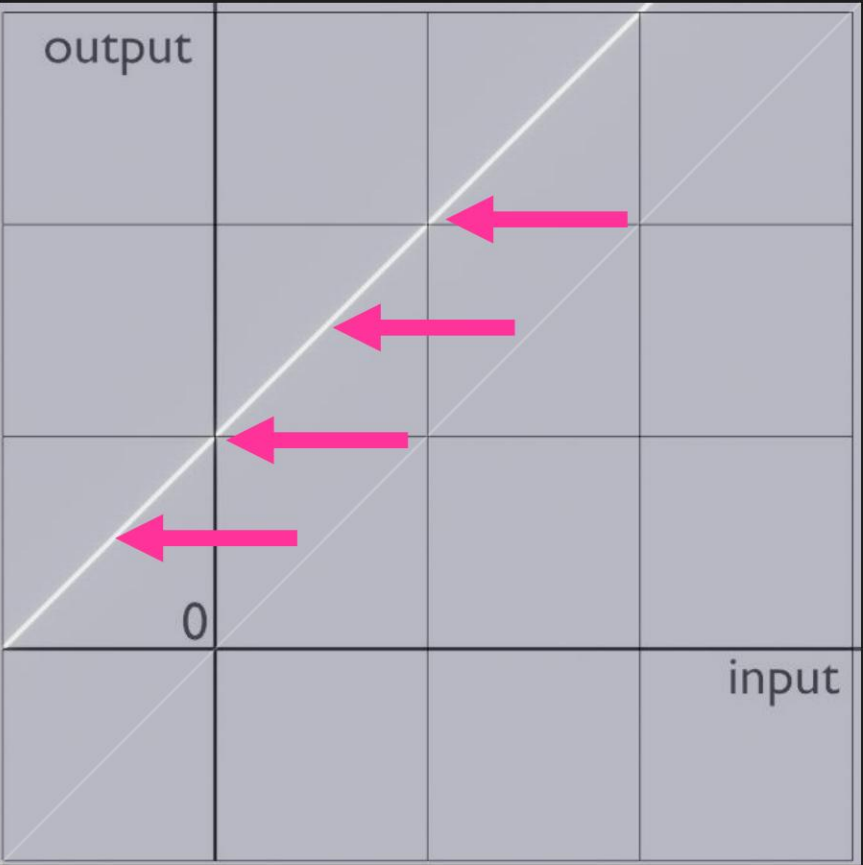
principle of movement

Adding 1 to each point...



principle of movement

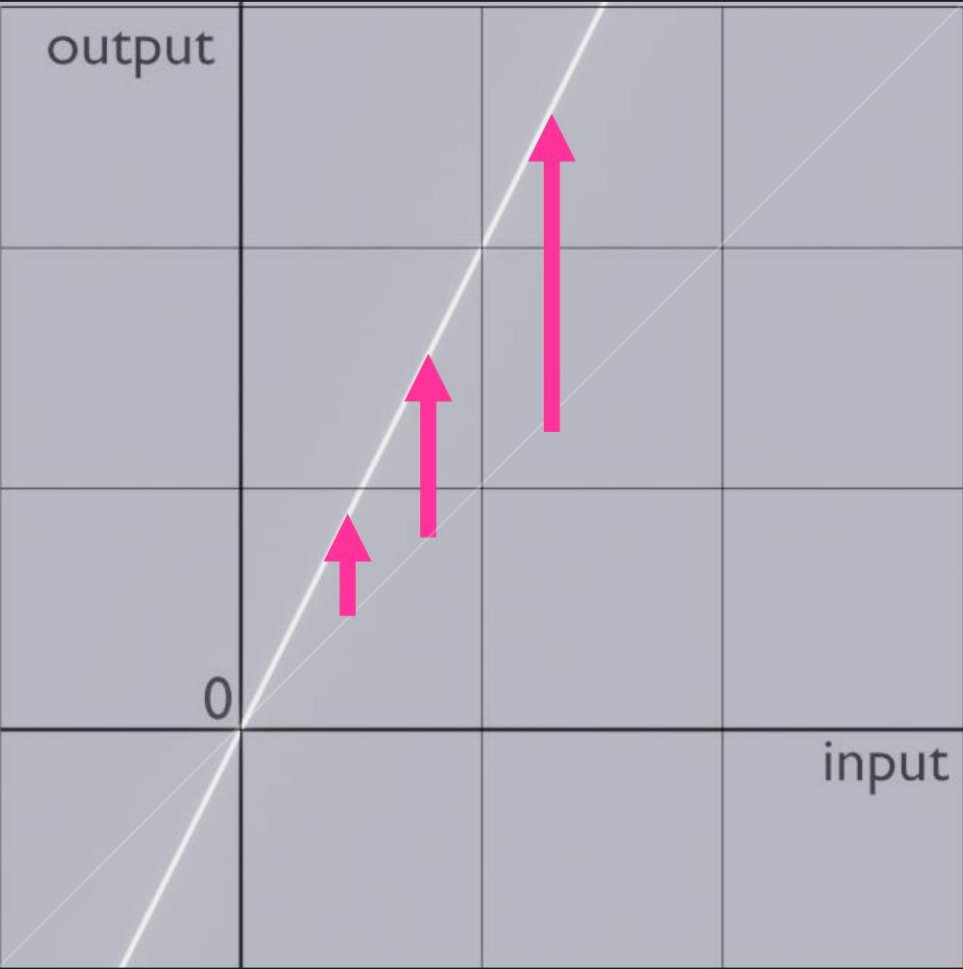
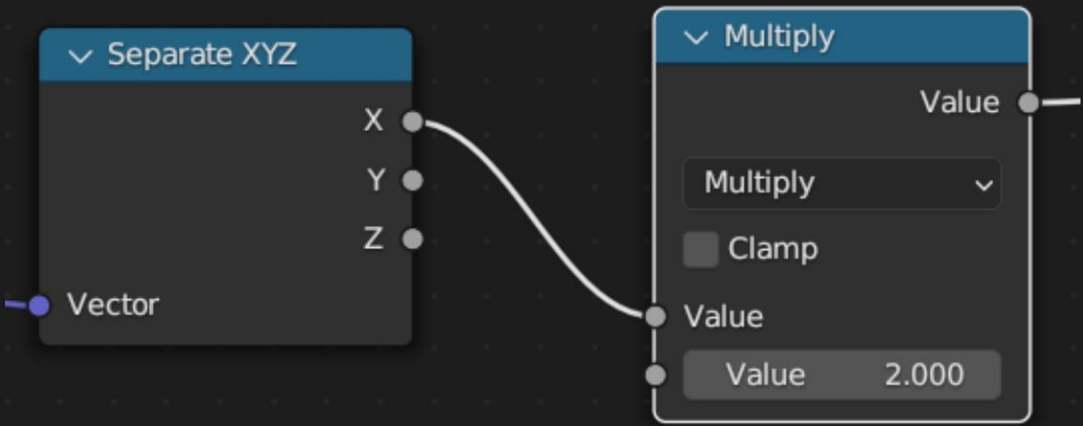
This effect moves the graph by 1 to the left.



3D is hard to imagine in your head, but the principle is the same as in 2D.

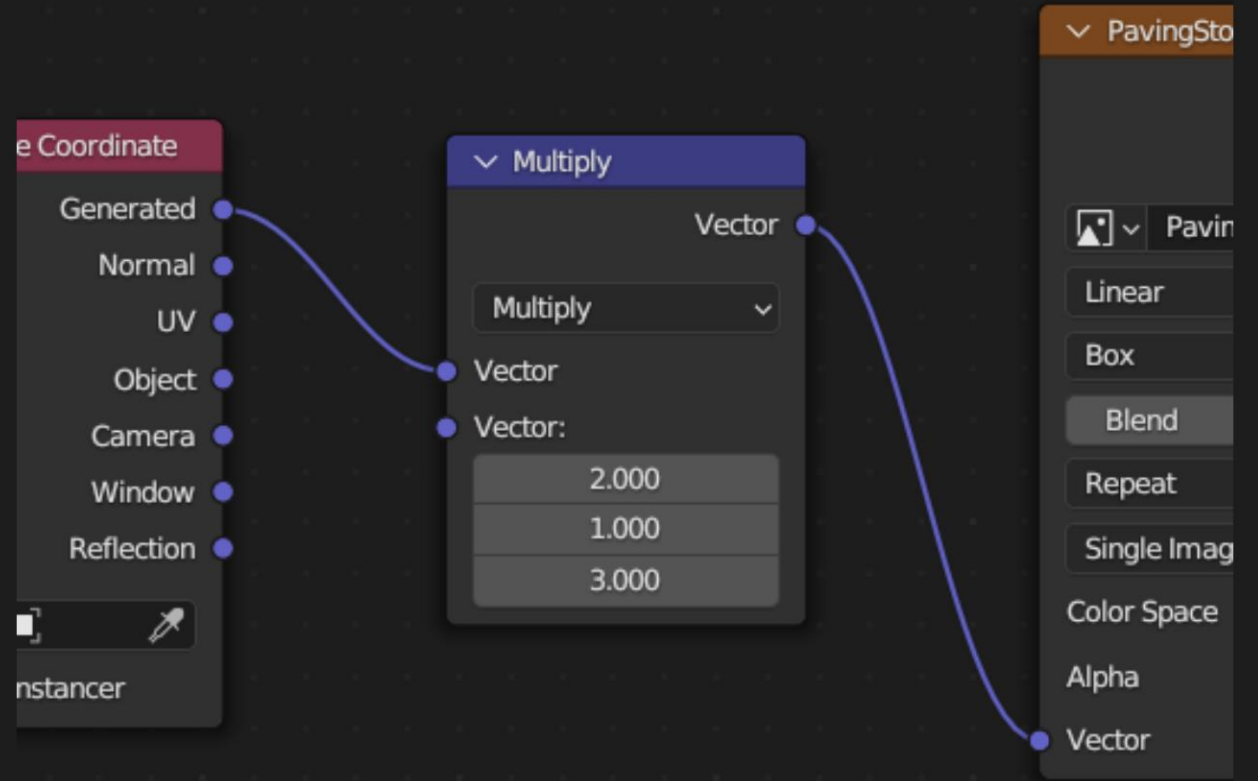
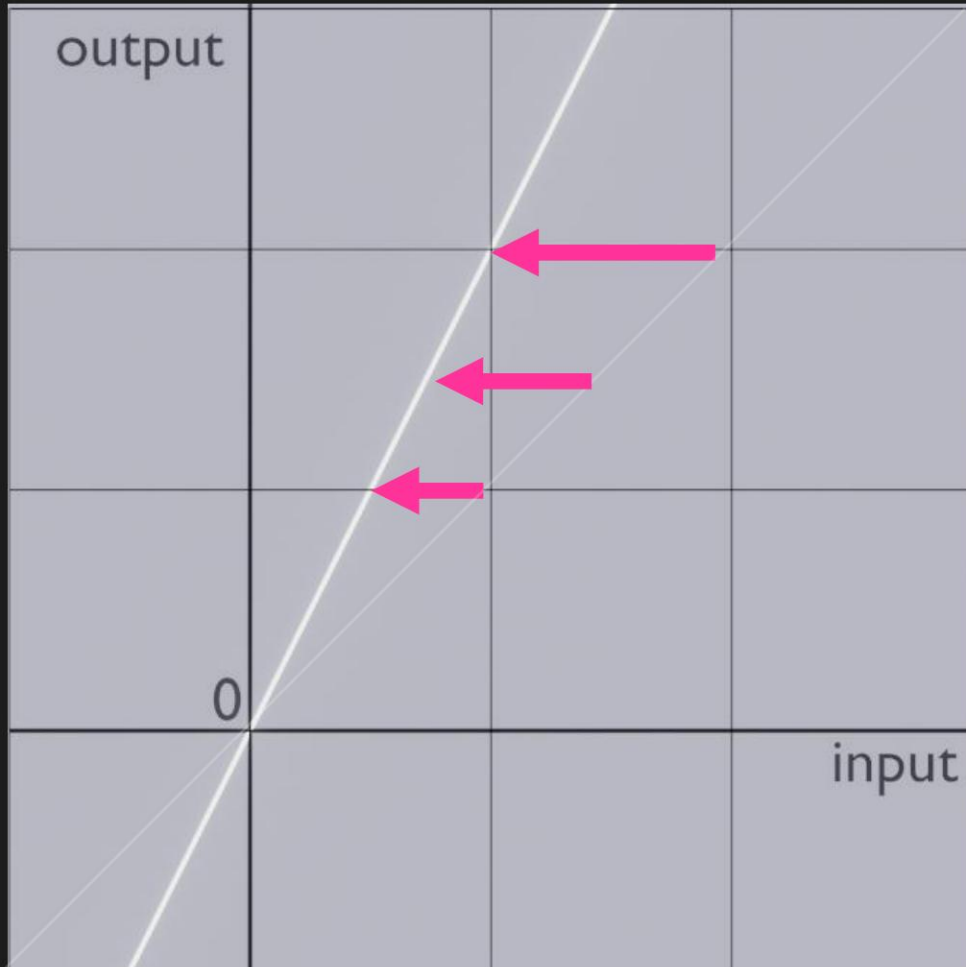
Principle of zooming

Multiply by 2 to increase the height of the graph...



Principle of zooming

It's the same effect as squeezing the graph.



Multiply by a large value to speed up the change. That is, it is scaled down.

Equation of Rotation?

$$\ddot{y} \cos \ddot{y} \ddot{y} \sin \ddot{y}$$

$\ddot{y} \sin \ddot{y} + \cos \ddot{y}$ (this is a 2D rotation. 3D rotation is more complicated)

Mapping

Vector

Type: Point

Vector

Location:

X	0 m
Y	0 m
Z	0 m

Rotation:

X	0°
Y	0°
Z	0°

Scale:

X	1.000
Y	1.000
Z	1.000

checkpoint

-Operations of shader nodes are done individually for each point.

In other words, passing data from one point to another is impossible!

-The meaning of addition and multiplication is a concept beyond the concept of coordinates.

In other words, the meaning of movement of addition and the meaning of enlargement/reduction of multiplication can be used universally.
It's just that the meaning of 'move' changes depending on the situation.

Rotation and Box Mapping

Box Mapping is based on Bounding Box.

However, even if you rotate the texture coordinates with the mapping node, the Bounding Box does not rotate. Because of this discrepancy, rotation does not work well for box mapping of 2D images.

Mix Shader



Blend the two shaders.

The lower the factor is, the higher it is, and the lower it is.

005 How to use the Mix Node

Shader Node Shortcuts and Gestures

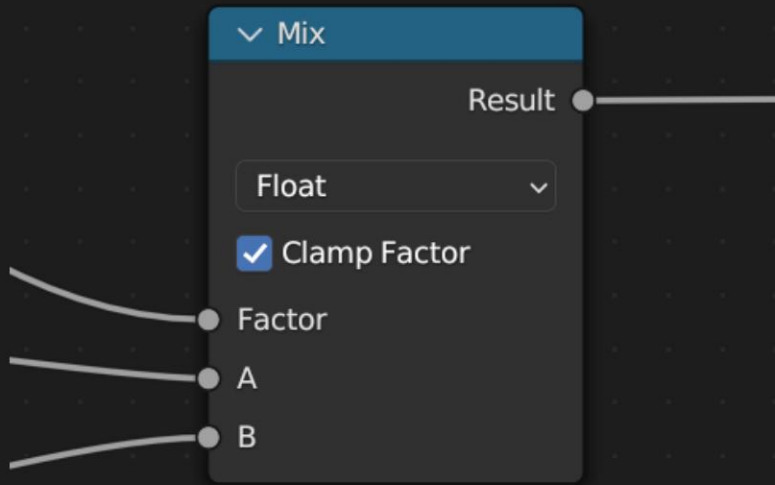
How to use each blending mode of the Mix node

Node editor shortcuts and gestures

Shortcuts and Gestures	explanation
N	Context menu
Shift+A	Create node menu
Ctrl+X	delete without disconnecting
Alt+drag	pull out without breaking connection
m	off node
Shift+D	duplicate node
Ctrl+Shift+D	Duplicate keeping connections
Ctrl+Shift+Click	View in Viewer
Ctrl+T	Link Texture
Ctrl+ right mouse button drag	Disconnect Cut off
CTRL+ALT+ RIGHT- DRAG	CONNECTION 'OFF' 禁用节点链接
Shift+ Right -drag	Make Knots 分割节点
Alt+ right mouse drag	Lazy Connect
Shift+Alt+ right mouse drag	Connect Shift + Alt+ Right Drag 选择链接到...
Ctrl+Shift+ Right Drag	Lazy Mix 快速混合

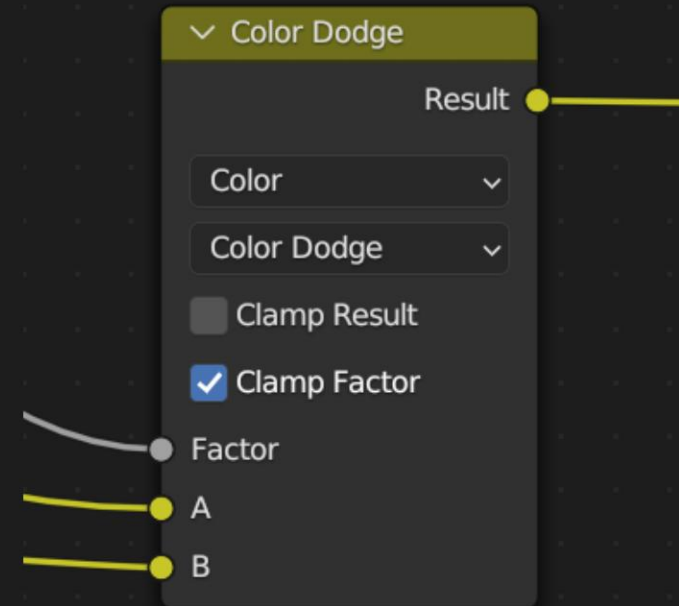
Shortcuts and Gestures	explanation
Ctrl+G	Create group
Ctrl+J	Create Frame
Alt+P	Export from node frame
Name	F2
Not Shortcuts, But Useful Things	
Ctrl+C, Ctrl+V	Copying/pasting of values is possible only by hovering the mouse over the values without dragging them.
+, -, *, /	in numerical input Calculations can be made in numerical input.
注: tau 常数值 数学上表示圆周率π (pi) 的两倍, 即 $\tau = 2\pi$ pi, e, tau pi, and natural constants can be entered for numerical input.	
Input beyond the slider	By entering directly, you can enter values beyond the slider.
Top Menu View - Auto Offset On/Off	the function that automatically widens the distance
Go to the top menu magnet button grid	

Mix node Linear Interpolation, or Layer Blending Mode



A weighted average between A and B, or Linear Interpolation. (Lerp)

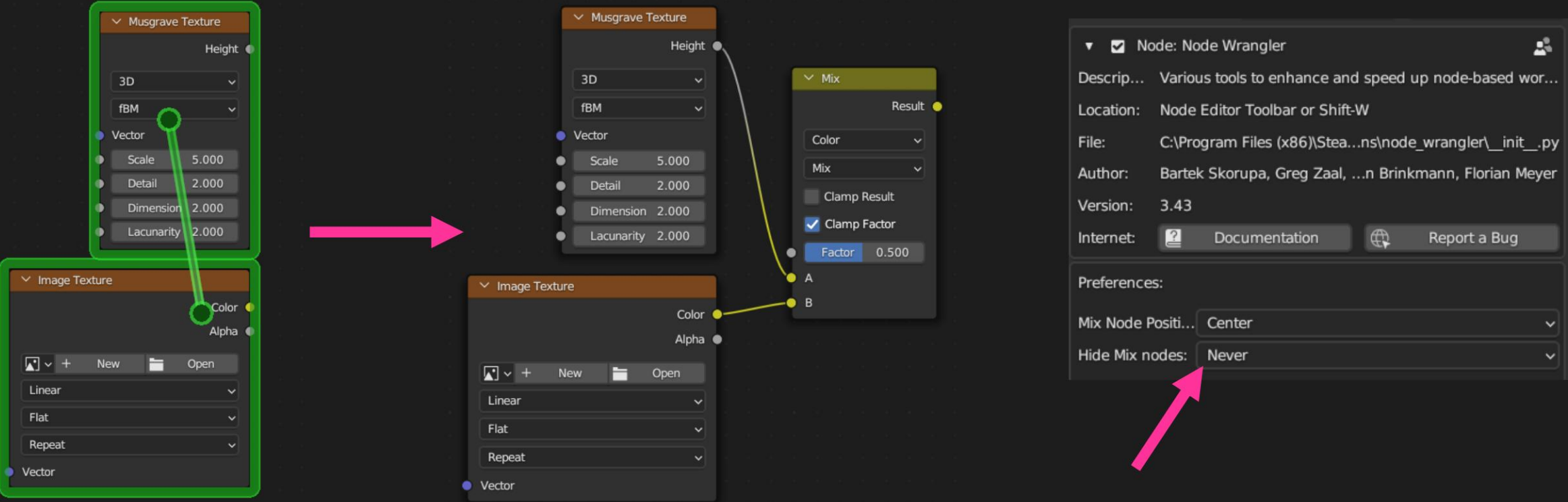
$$(1 - \text{Factor}) \cdot A + \text{Factor} \cdot B$$



In case of color mode, blending mode is also supported.

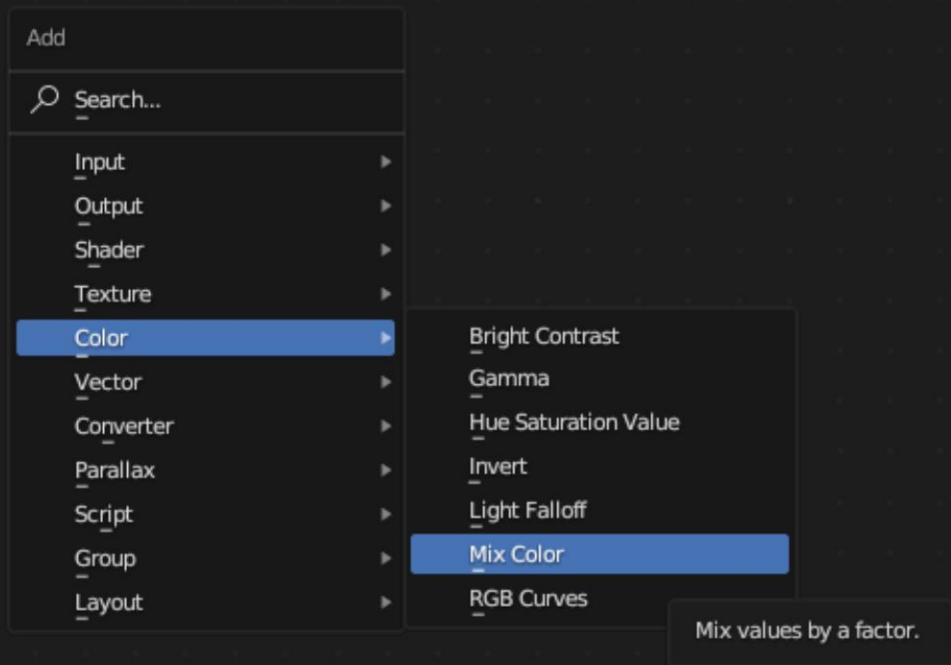
Mix node quick access

Lazy Mix: Ctrl+Shift+ **right** mouse button drag

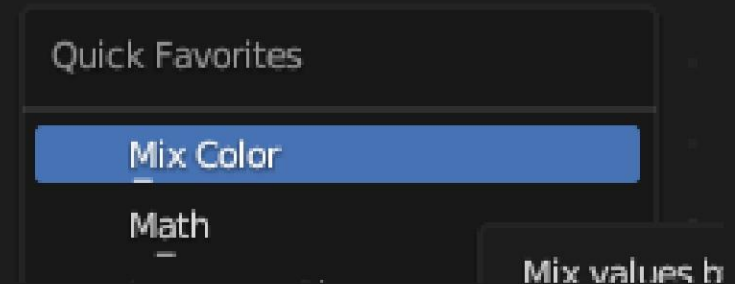


Mix node quick access

Mix Color creates a Mix node in Color mode.



... .. If either method is inconvenient, put it in your Quick Favorites.



Mix node

▼ Mix

Result

Color

Mix

Mix

Darken

Multiply

Color Burn

Lighten

Screen

Color Dodge

Add

Overlay

Soft Light

Linear Light

Difference

Subtract

Divide

Hue

Saturation

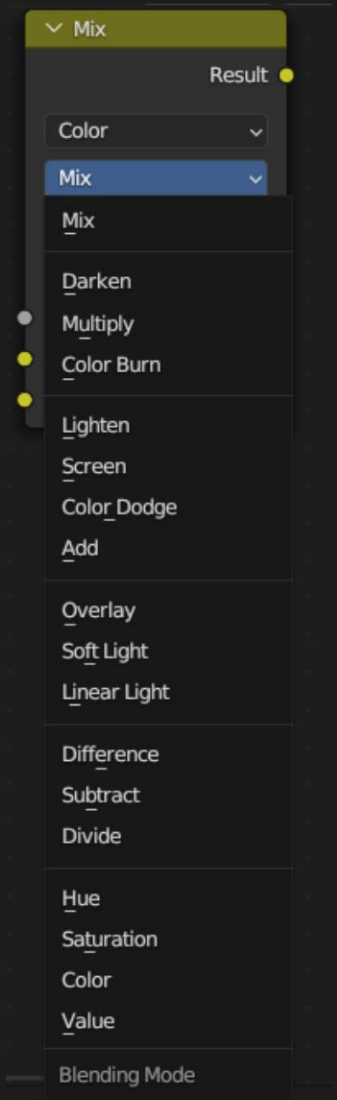
Color

Value

Blending Mode

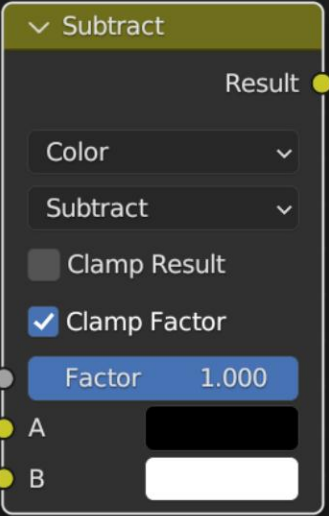
	Blending Mode	ceremony	explanation
darkly	Darken	$\min(A,B)$	Outputs the darker of the two.
	Multiply	$A \times B$	It becomes darker as if you raised the tone of B on top of A.
	Color Burn	$1 - (1-A)/B$	Darken A with B. Brightness is lowered and saturation is increased.
brightly	Light	$\max(A,B)$	Outputs the lighter of the two.
	Screen	$1 - (1-B)(1-A)$	Contrary to Multiply , it brightens as if you shot B over A onto a screen.
	Color Dodge	$A/(1-B)$	Use B to light up A. Brightness goes up and saturation goes down.
	Add	$A+B$	Simple sum of A and B. As a result, it brightens.
contrast	Overlay	$(\text{When } A \geq 0.5) \ 1 - 2(1-A)(1-B)$ $(\text{When } A < 0.5) \ 2AB$	It works similarly to Screen when A is bright and Multiply when it is dark .
	Soft Light	$(1-2B)A^2+2BA$	Blends more gently than Overlay .
	Linear Light	$A+2B-1$	Add the range of B to A by extending it $(=2B-1)$. Mix stronger than Overlay .

Mix node

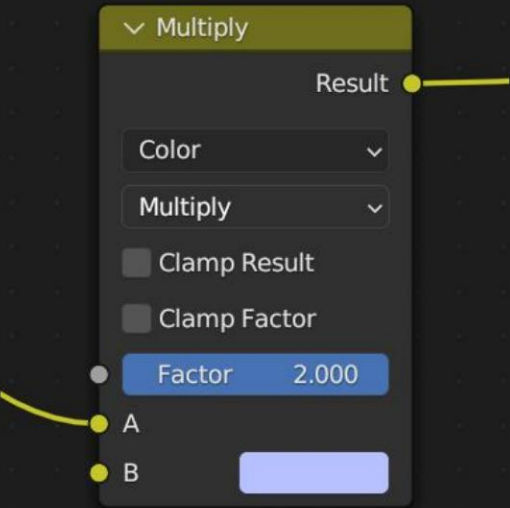


	Blending Mode	ceremony	explanation
reversal	Difference $ A-B $		Difference between A and B (positive number)
	Subtract $A-B$		difference between A and B.
	Exclusion $A+B-2AB$		Invert A by B
	Divide A/B		Divide A by B. It works the 'mathematically' opposite of Multiply . If Multiply tones the image, Divide removes the tone from the image.
color	Hue $HSV(H(B),S(A),V(A))$	Use	B's color for image in A.
	Saturation $HSV(H(A),S(B),V(A))$	Uses	the saturation of B for the image of A.
	Color $HSV(H(B),S(B),V(A))$	Uses	the color and saturation of B for the image in A.
	Value $HSV(H(A),S(A),V(B))$	Use	B brightness for image A.

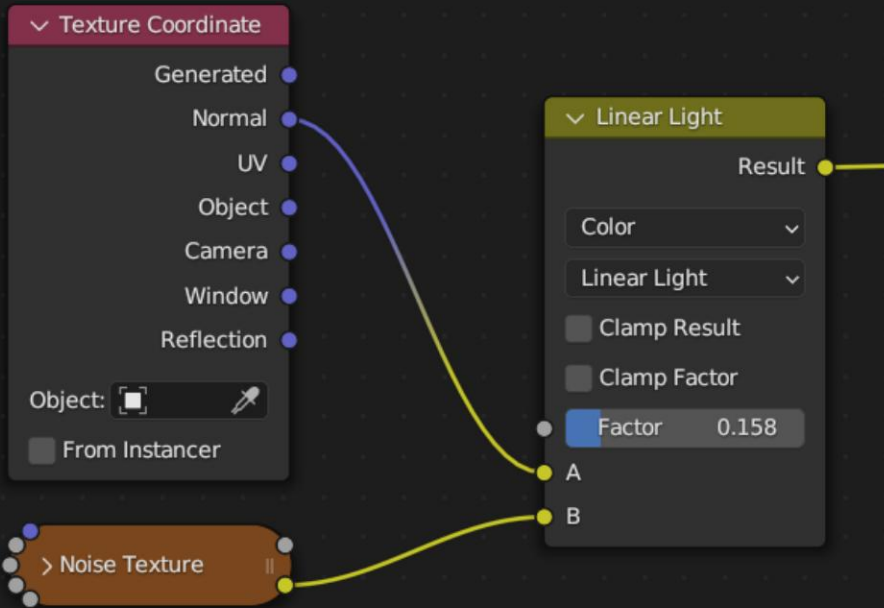
This is not an illustration program, more free.



Color values also drop to negative numbers



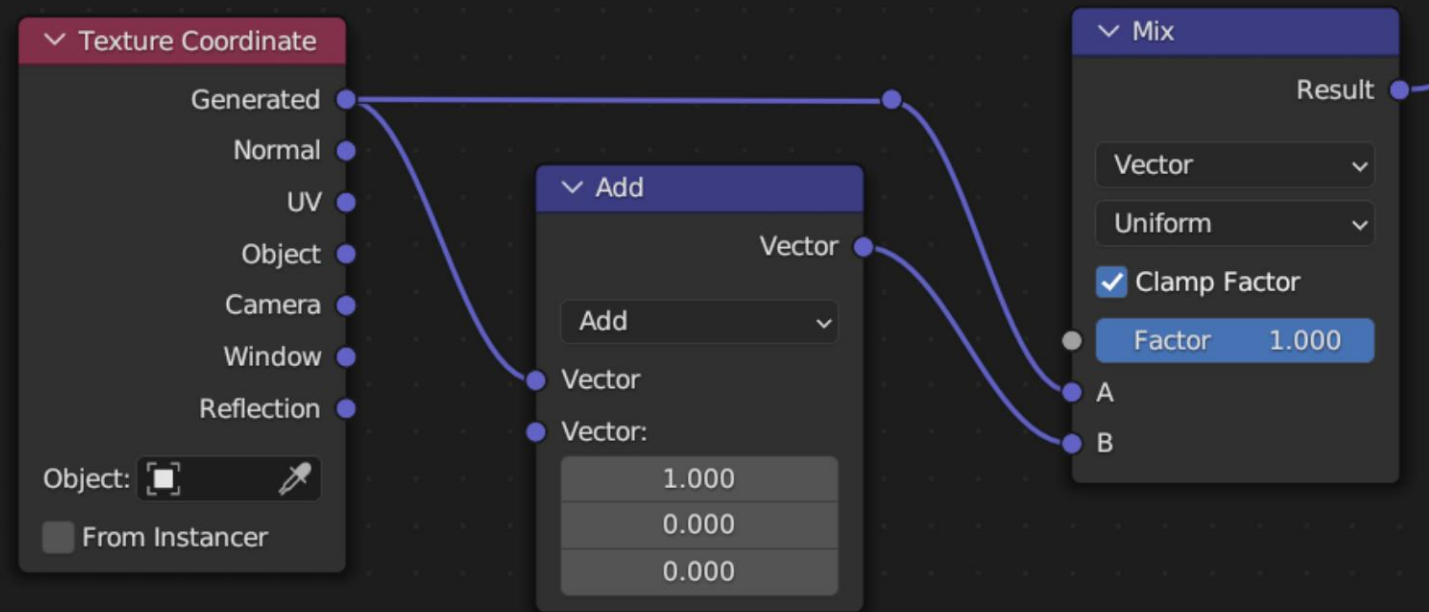
You can multiply by
'double' by adjusting the factor



'Linear Light' of vectors can also be calculated

Use Clamp if you do not want results that exceed the range.

This is not an illustrator program



You don't have to imagine the result as an image.

006 Compilation of images 图像编辑

How to edit images like Photoshop
Understanding Gamma Correction
and Filmic Filter Creating Faded Photos Using Nodes

电影滤镜 褪色照片效果



not photoshop

Separate Color RGB 本质与 Separate XYZ 相同

▼ Hue Saturation Value

Color

Hue

0.500

Saturation

1.000

< Value

1.000

>

Fac

1.000

Color

▼ Invert

Color

Fac

1.000

Color

▼ RGB Curves

Color

C

R

G

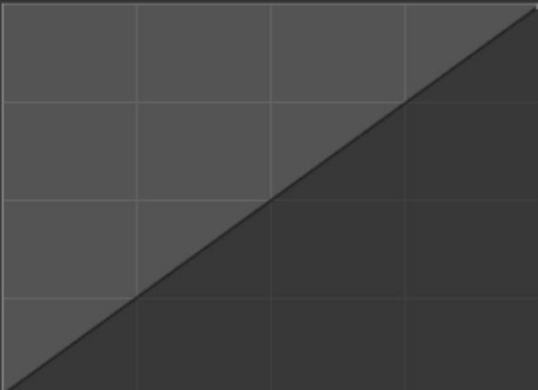
B

+

-

●

▼



Fac

1.000

Color

▼ Bright/Contrast

Color

Color

Bright

0.000

Contrast

0.000

▼ Gamma

Color

Color

Gamma

1.000

▼ RGB to BW

Val

Color

▼ Separate Color

Hue

Saturation

Value

HSV

▼

Color

▼ Combine Color

Color

RGB

▼

Red

0.000

Green

0.000

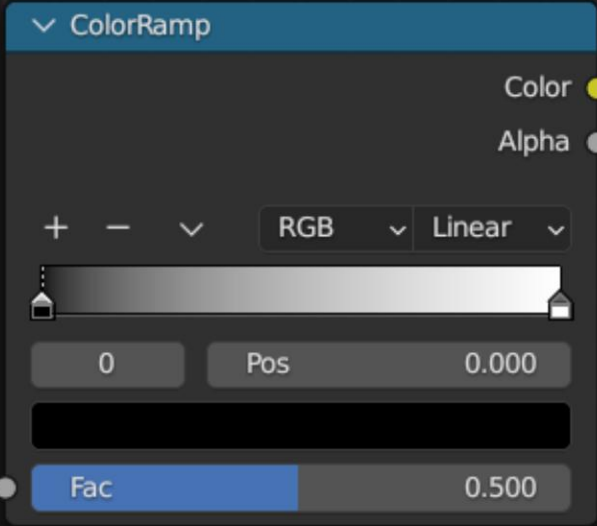
Blue

0.000

ColorRamp

在图形设计和渲染软件中，“ramp”这个词经常被用来指代“梯度图”或“渐变”。梯度图是一种颜色、亮度或任何其他视觉属性的渐变表示，通常用于创建平滑的颜色过渡效果。

对于黑白色(灰度信息(值))的输入, Color ramp 采用颜色映射, color ramp将灰度值映射到定义的颜色上。例如, 较暗的灰度值(接近黑色)可以被映射到锚点定义的起始颜色, 而较亮的灰度值(接近白色)可以被映射到结束颜色。



Corresponds colors according to the entered value (Factor). The default interpolation method is Linear, but smoother transitions are also possible using B-Spline. (This is covered in detail in the [Procedural Texture](#) section.)

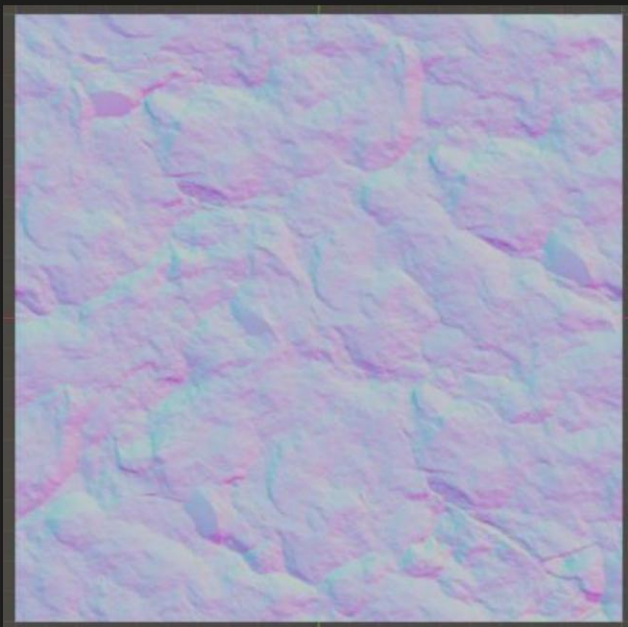
后续的[程序化纹理](#)章节会提到 B-Spline

=
在仅需要黑白图像的情况下, 左右俩节点功能相同

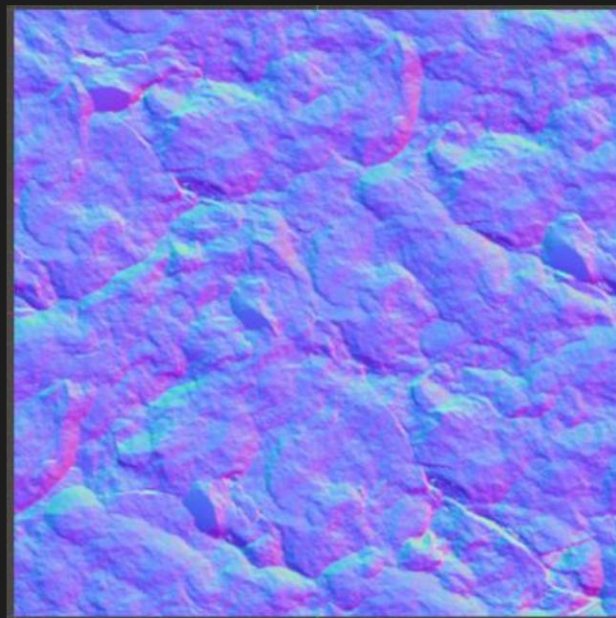


If only black and white images are needed, Float Curve also plays the same role.

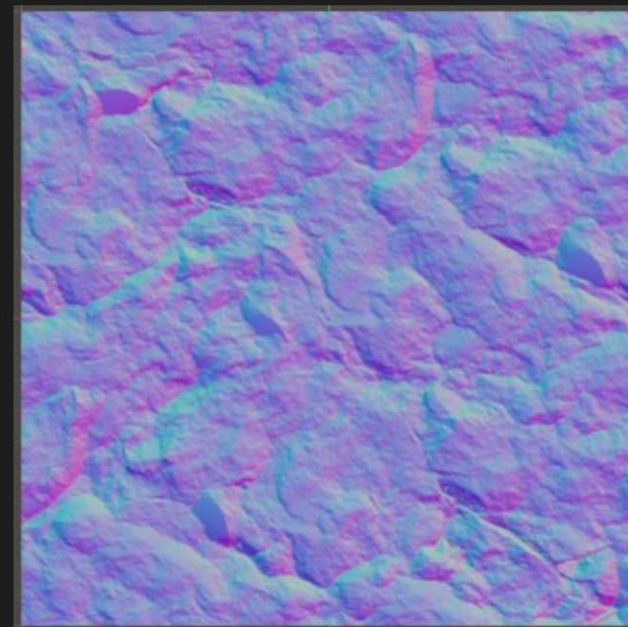
color problem



gamma correction

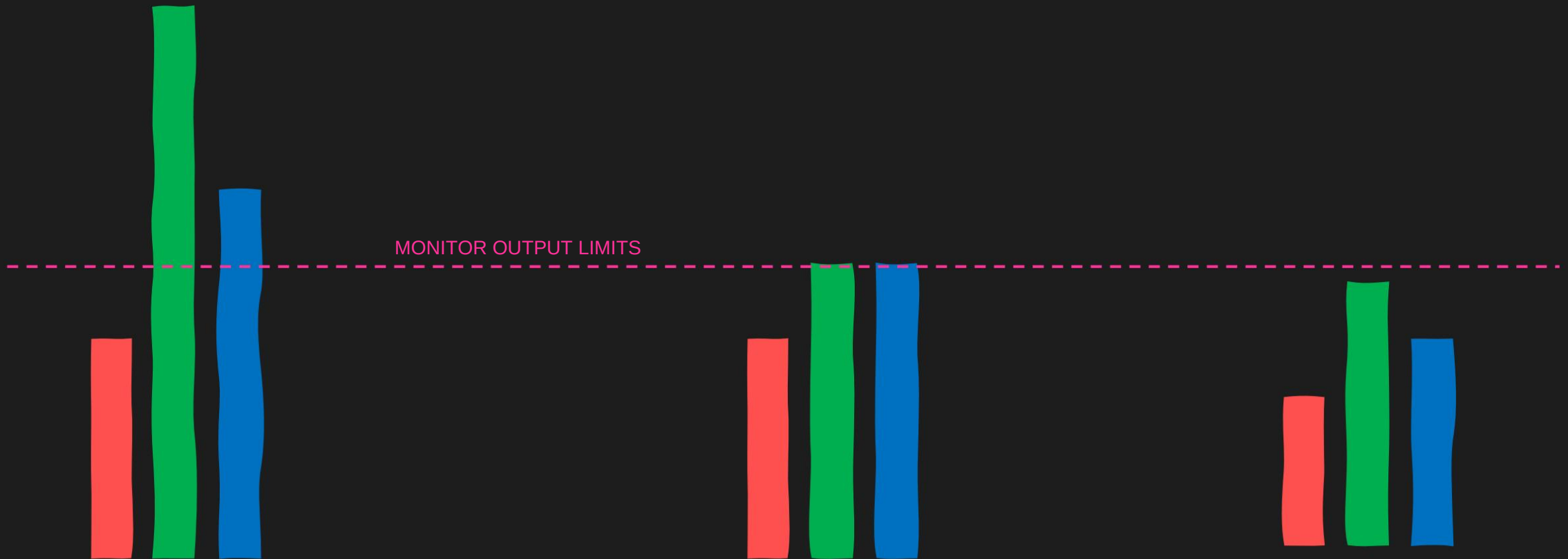


sRGB



Filmic Filter

Filmic View Transform



When Blender creates an image, it creates colors without constraints on brightness.

If printed as it is, a value greater than 1 is indistinguishable.

For a natural image, a full color transformation is required.

Filmic View Transform

Before Applying View Transform



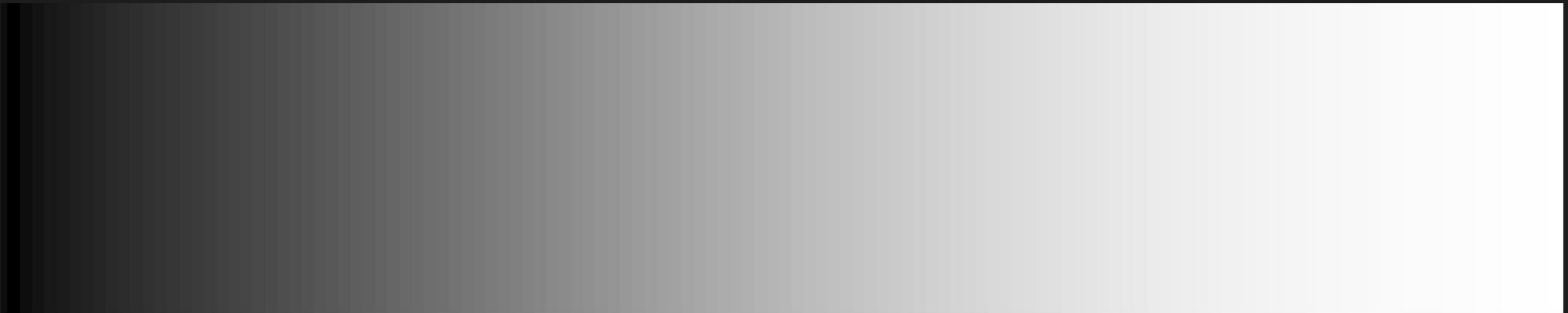
After applying View Transform



Correspondingly, images with a brightness between 0 and 1 will also change color, subtle but different from the original. (It may feel cloudy or cloudy.)

gamma correction

Human eyes are more sensitive to dark areas. Even when saving an image, it stores more information in dark areas.



Therefore, when displaying an image on the screen, the stored value is not displayed as it is, but displayed after **some calculation**. That's gamma correction.

gamma correction

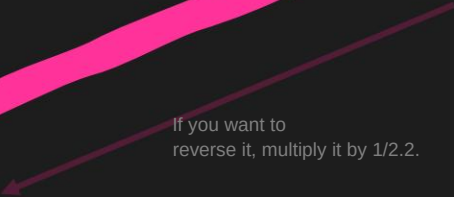


image source data

Gamma correction (power of 2.2)



If you want to reverse it, multiply it by $1/2.2$.

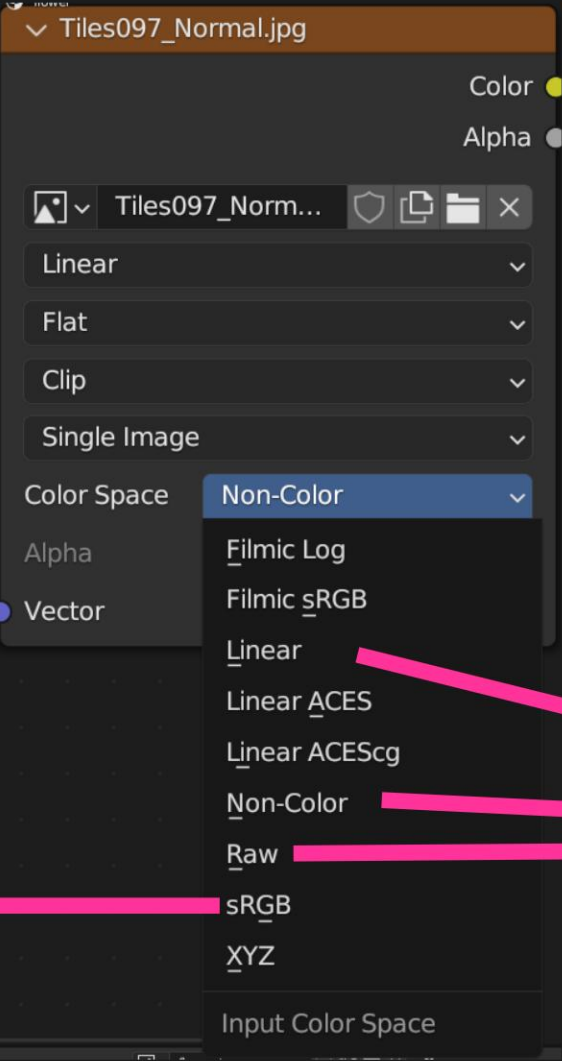


Output image of a typical monitor



Substituting the original data as it is for color

gamma correction

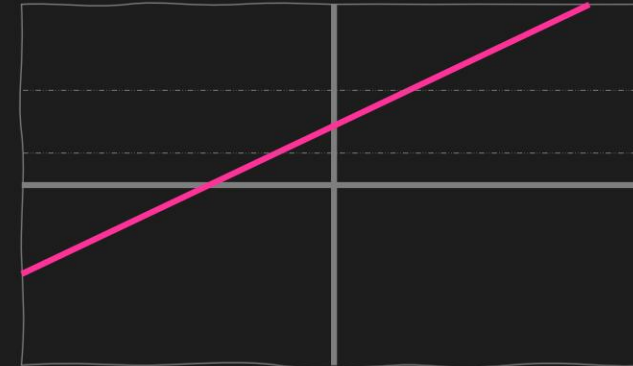
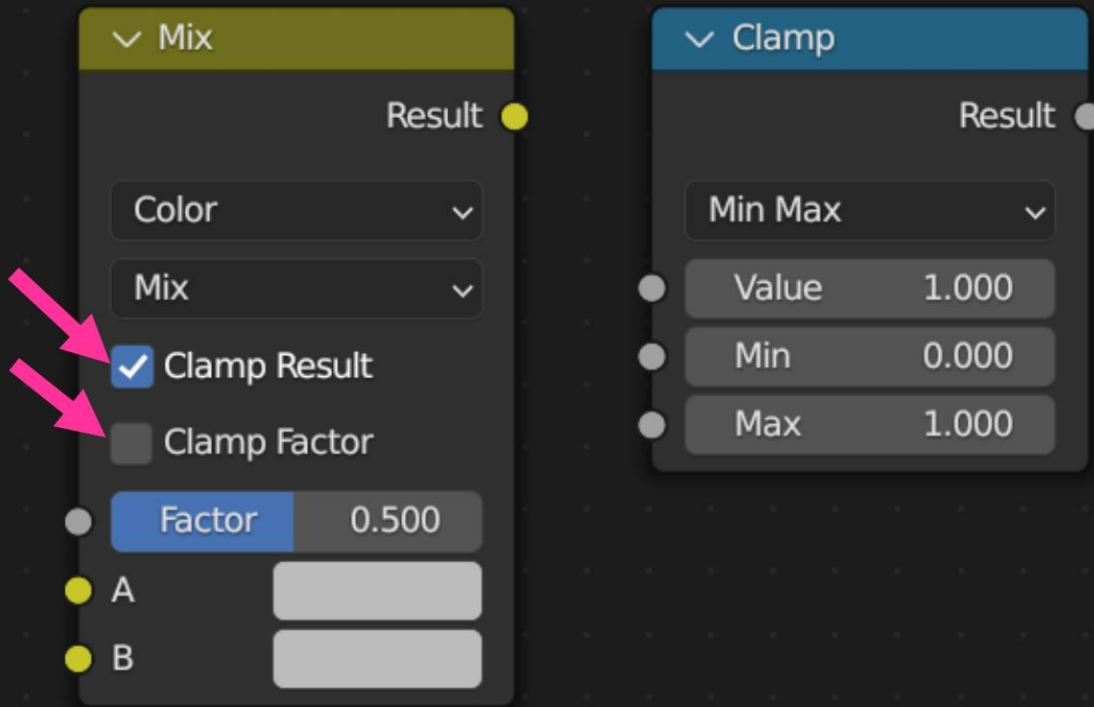


The default value,
sRGB, is gamma-corrected (to the power of 2.2).

If you want to use the normal
map, you must use the value before gamma correction.
Use **Linear**,
Raw,
or **Non-Color** (they are all the
same values).

Clipping problem

In general, the result of an operation is not limited by a maximum or minimum value. Direct control as needed.



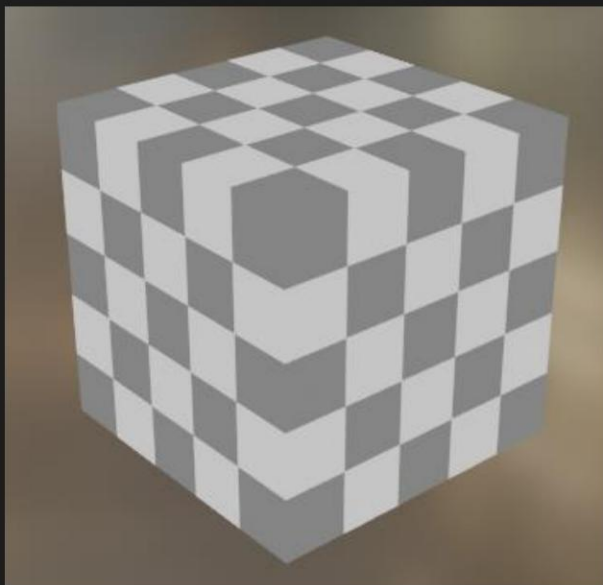
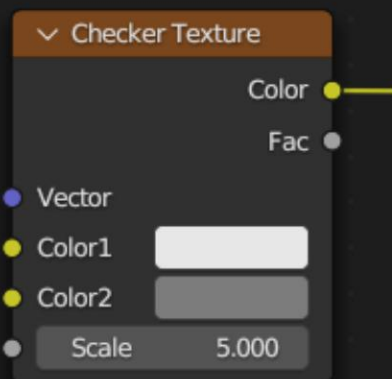
007 built-in texture

-Procedural textures built into
Blender -Let's make a cloth material

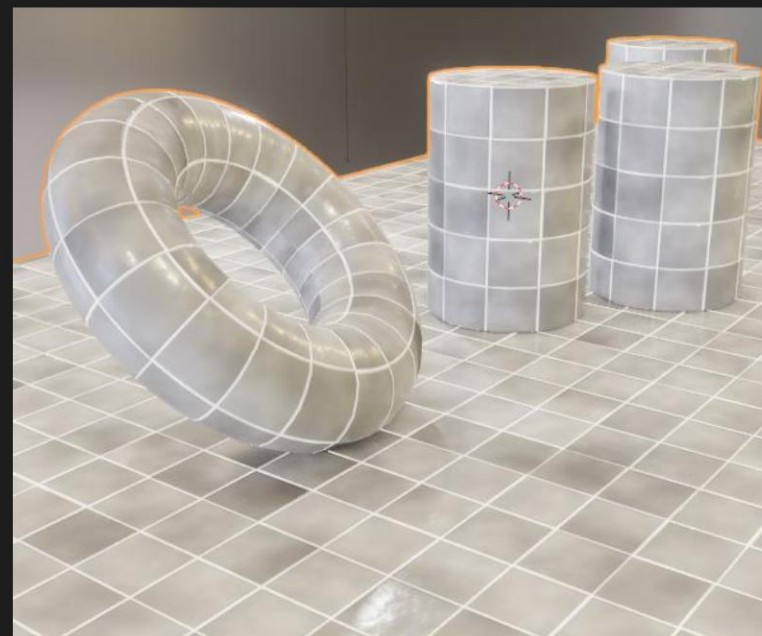
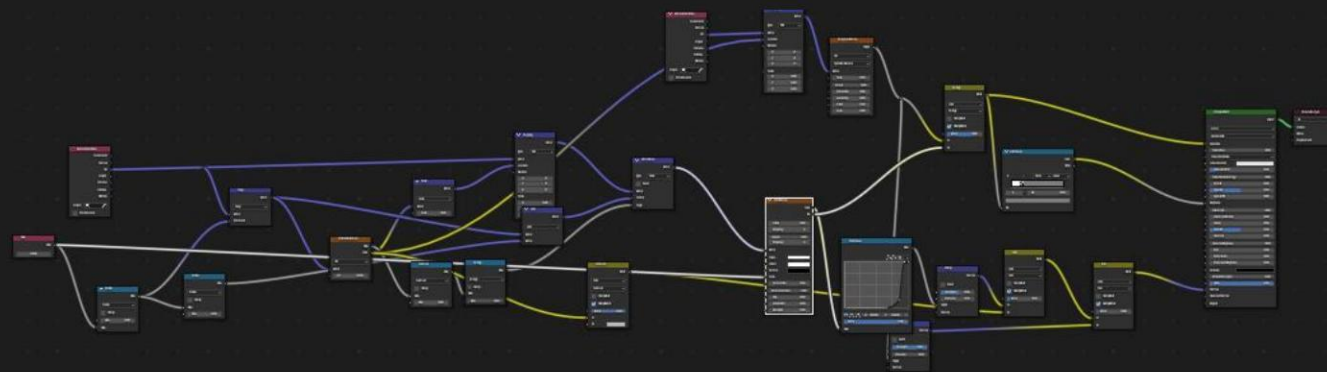


Procedural Texture - A texture created using a mathematical method.

Blender Basic Textures



... Or a texture made by connecting them

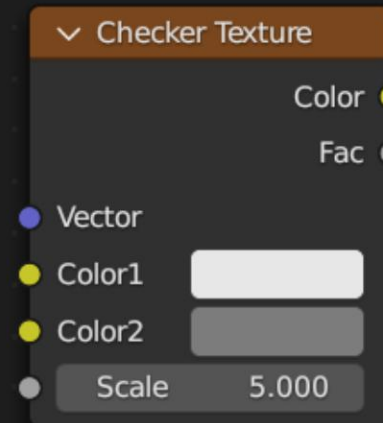


Brick Texture, Checker Texture



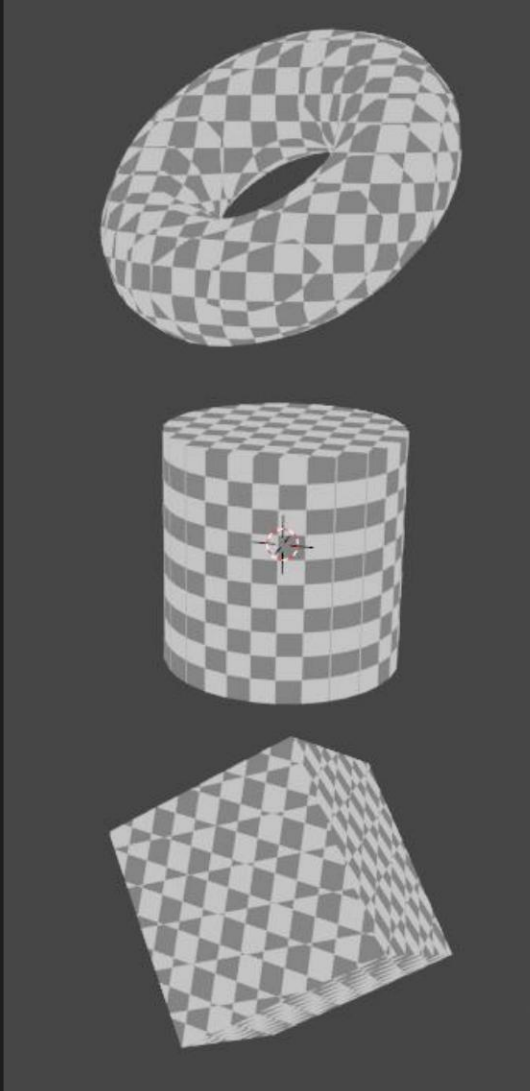
Create a brick pattern.

You can adjust the pattern with Offset and Squash.



Create a checkered pattern.

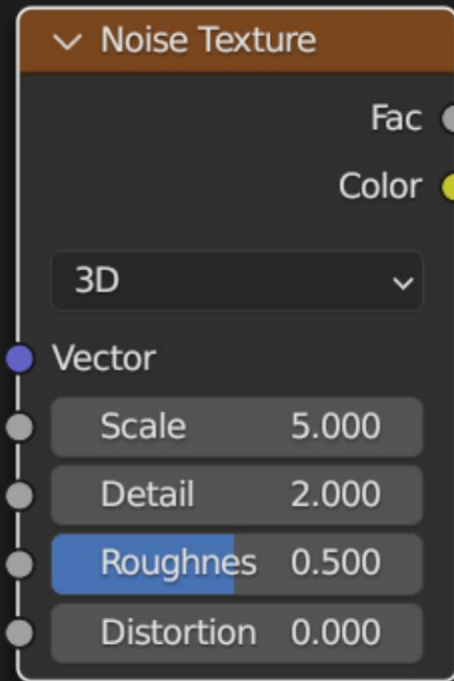
This is a 3D texture



Use all three dimensions of coordinates

Since it continuously exists in the 3D space, the seam that occurs when using UV – image texture does not occur. However, it can also be used in 2D if necessary.

Noise Texture



Detail: Complexity doubles for each increment of 1.

Roughness: Determines how much small noise patterns are blended.
When it becomes 0, Detail is ignored.

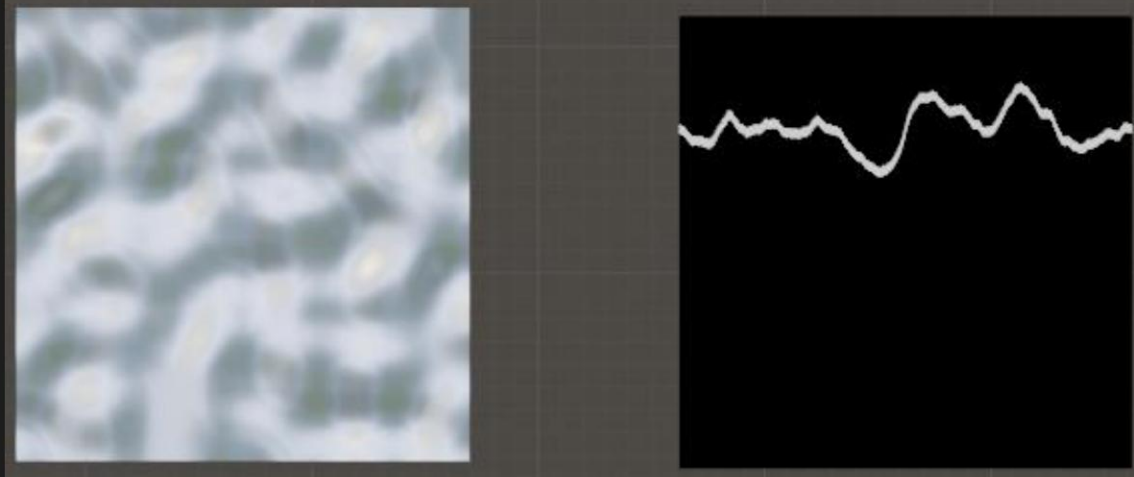
Several ways to visually check texture

image preview



Greater than

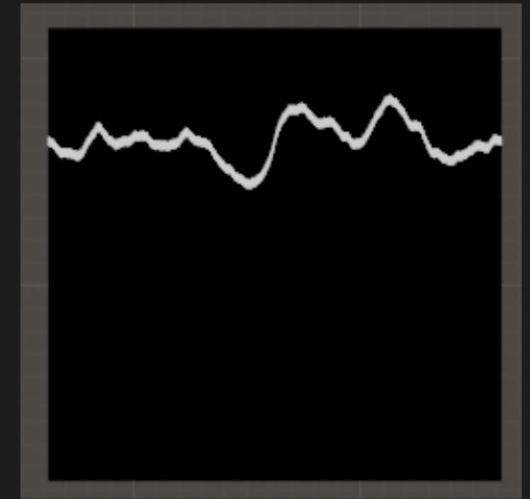
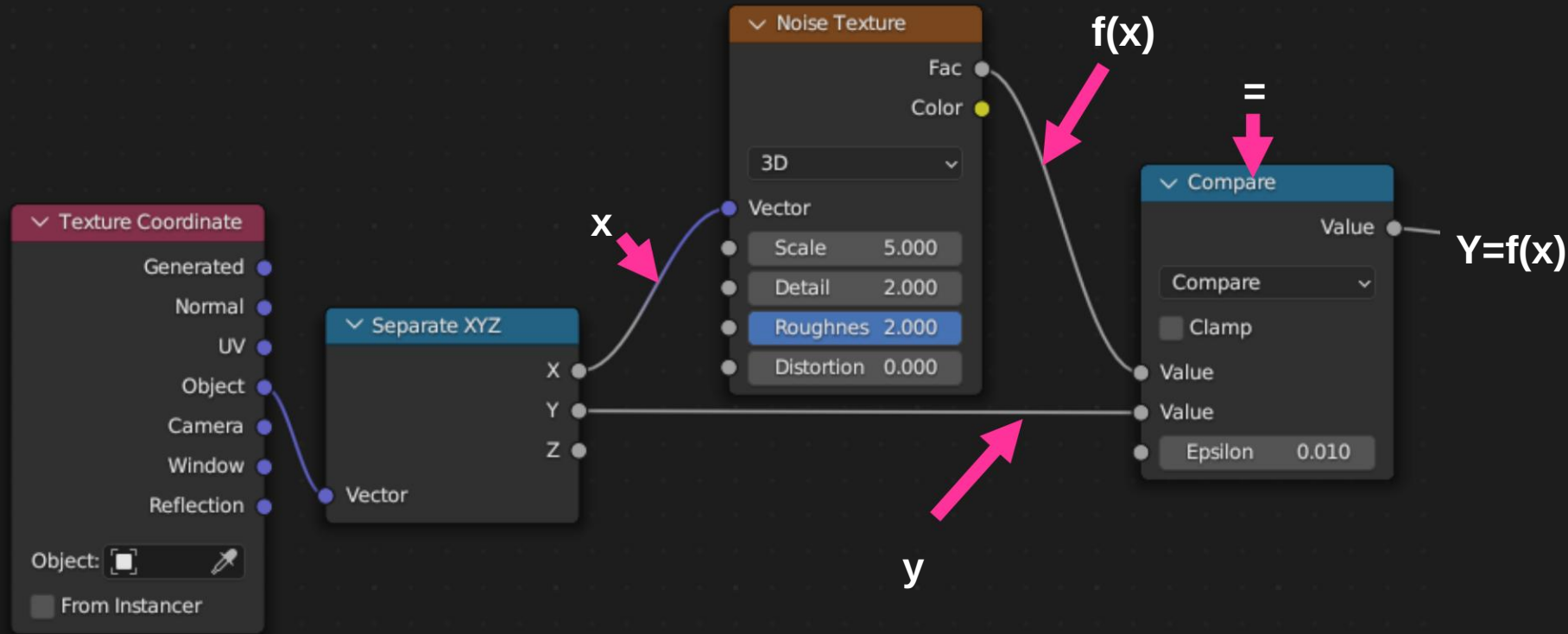
Bump



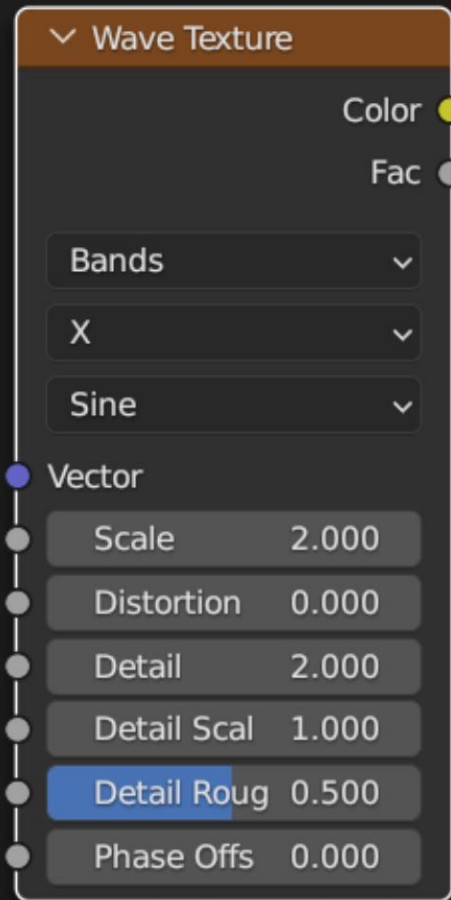
2D graph

Several ways to visually check texture

If you create $y=f(x)$ in Blender, you can see a 2D graph.

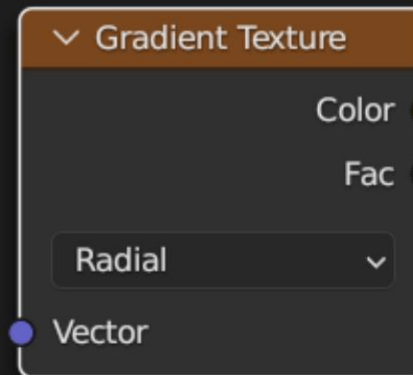


Wave Texture, Gradient Texture



Creates a periodic function.
 ↳ The range is always between 0 and 1.

Detail, Detail Scale, and Detail Roughness are all values for the noise Distortion creates.



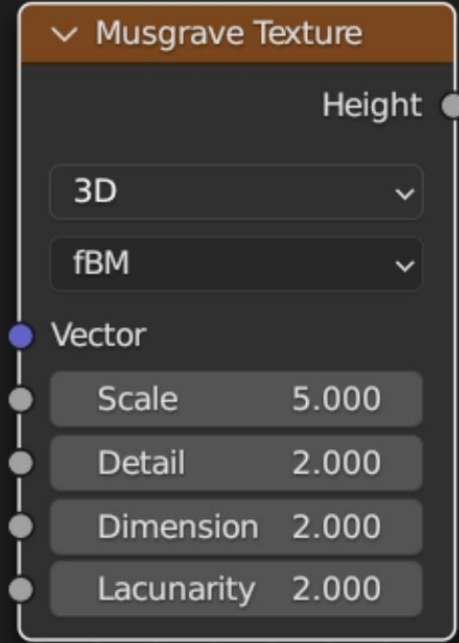
Create multi-shaped gradients.

↳ The range is clipped between 0 and 1.

↳ The center of Spherical, Quadratic Sphere, and Radial is (0,0,0). So when I use Object coordinates it appears fine.

Musgrave Texture (1)

Similar to noise texture, but...



Unlike noise, the output value is one-dimensional.

The range is 'generally' -1 to 1 , but goes beyond that as well.
In some cases, it is better to utilize the Map range node.

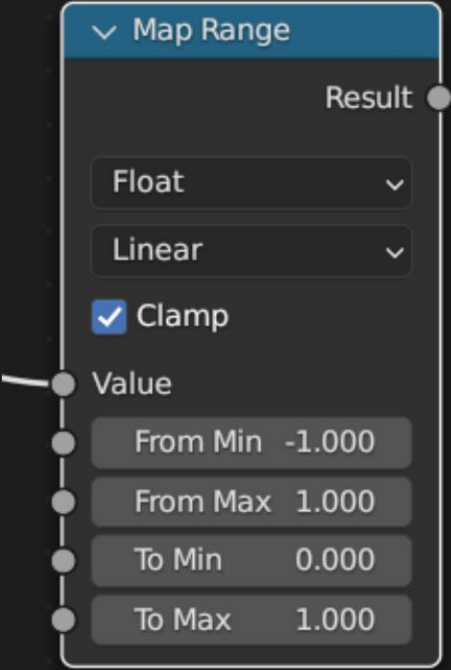
Detail has the same meaning as in the Noise texture,
but the way it controls the noise 'octave' is different.

Dimension: Adjusts the amplitude difference between octaves.
the exact formula is

$$= \frac{1}{x}$$

but..

Lacunarity: Adjusts the scale difference between octaves.



Musgrave Texture (2)^{effective combinations}

fBM



Dimension 1.4, Lacunarity 2.0

Multifractal



Dimension 1.4, Lacunarity 2.0

**Hybrid
Multifractal**



Dimension 0.1, Lacunarity 3.0
Offset 0.45, Gain 0.63

Rigid Multifractal



Dimension 0.1, Lacunarity 3.0
Offset 0.5, Gain 5.0



Dimension 0.1, Lacunarity 3.0
Offset -0.06 Gain 5.0

Hetero Terrains



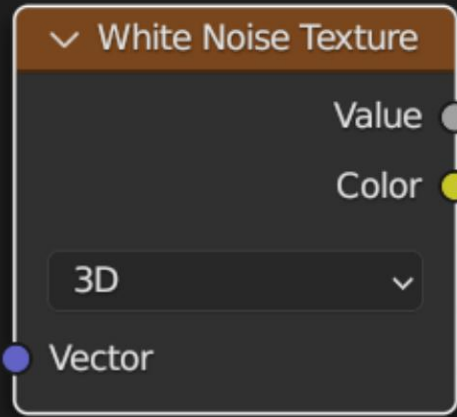
Dimension 1.0, Lacunarity 3.0
Offset 0.3



Dimension 1.0, Lacunarity 3.0
Offset -0.3

White Noise Texture

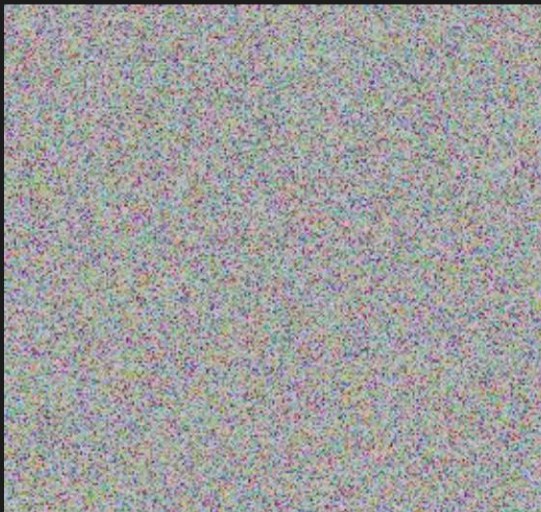
Is this... texture?



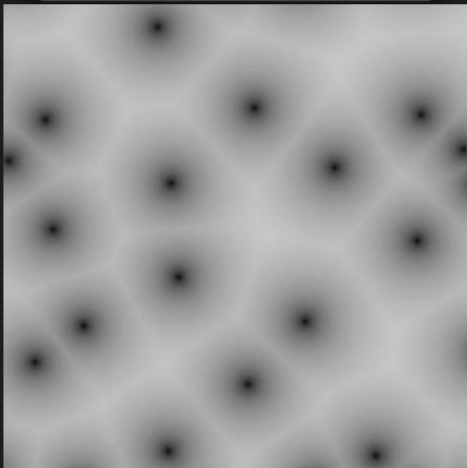
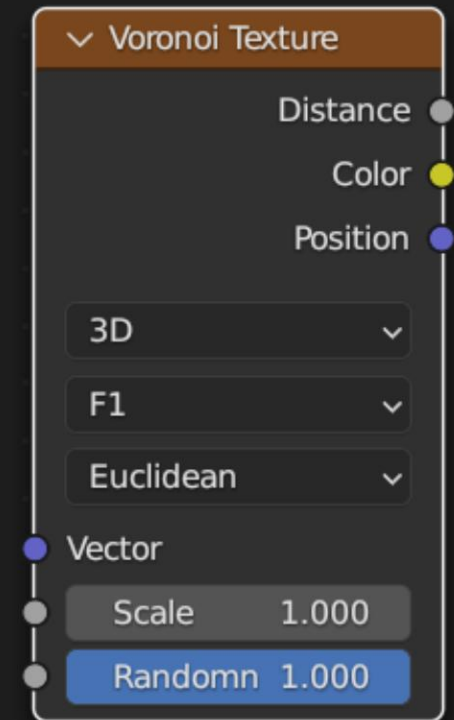
Outputs a random number between 0 and 1 depending on the input value.

Even if the input values are similar, completely different values are produced.

For example, for inputs 1.1 and 1.101, it gives completely different values of 0.943 and 0.792 respectively. It produces the same output only for **the same** input .



Voronoi Texture



Distance: Represents the distance from a point uniformly distributed in space.

A boundary is created based on which point is closest. Color output is a random color for each area that looks like that .

Position outputs the position value of each point.

It usually represents the distance to the closest point (F1), but you can change the options as below.

F2 : Outputs the distance from the 'second' nearest point.

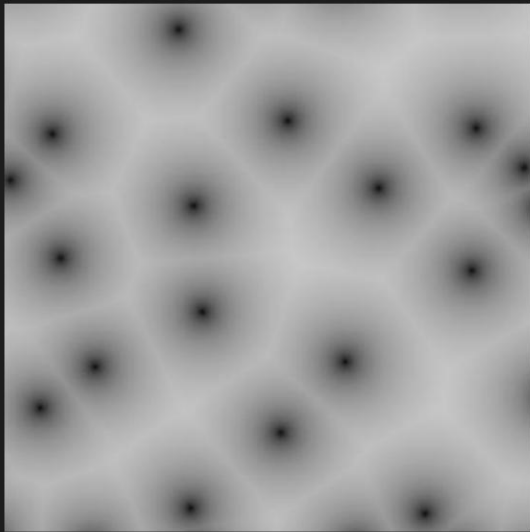
Smooth F1 : The boundary of the zone is softened.

Distance to Edge : Measures the distance 'from the edge', not the distance to the point.

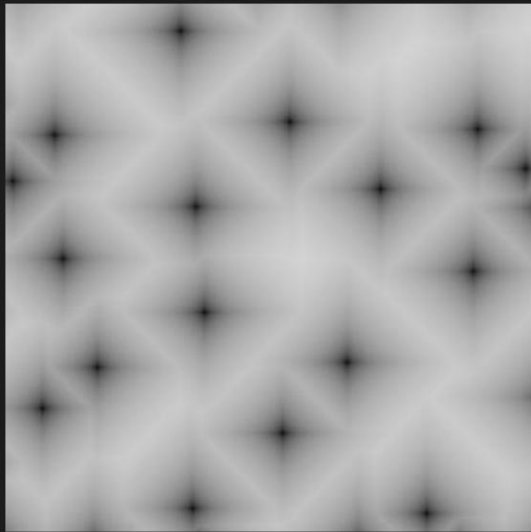
N Sphere Radius : By definition, it is the radius of a sphere centered at a point and inscribed on the boundary. In simple words, for each region, output the minimum distance from the point to the boundary. It doesn't work well.

Voronoi Textures (2)

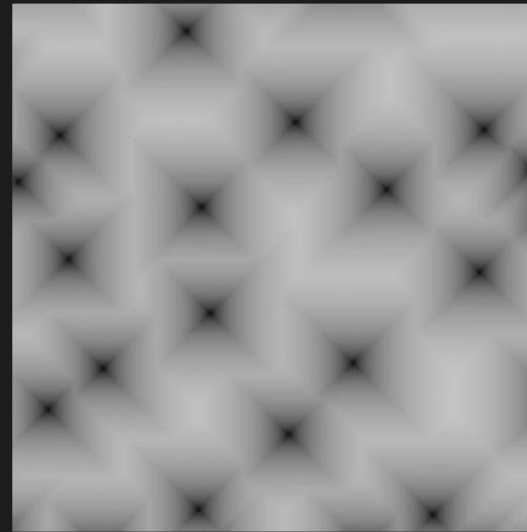
Euclidean, Manhattan, Chebyshev, Mincowski: change **how distances are measured**



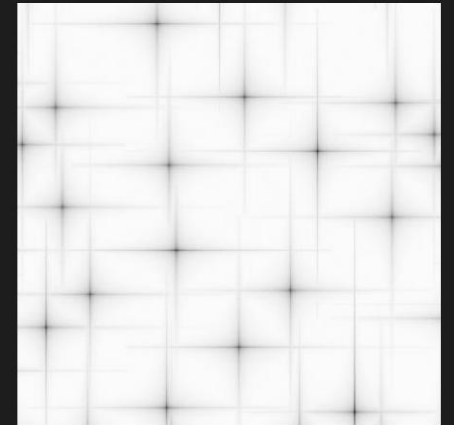
Euclidean



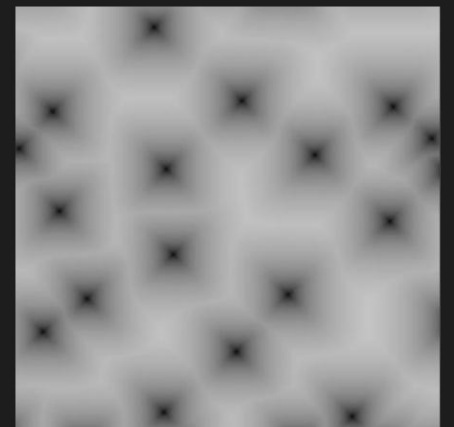
Manhattan



Chebyshev

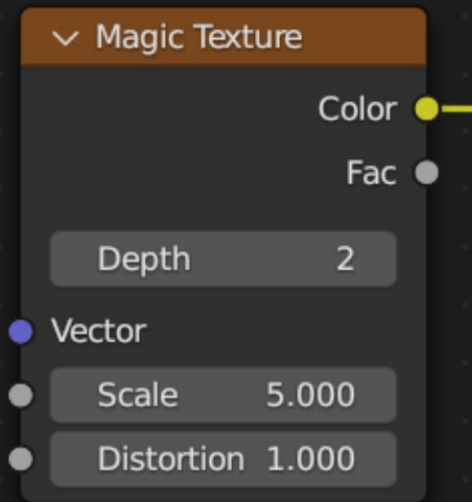


Mincowski ($e=0.3$)



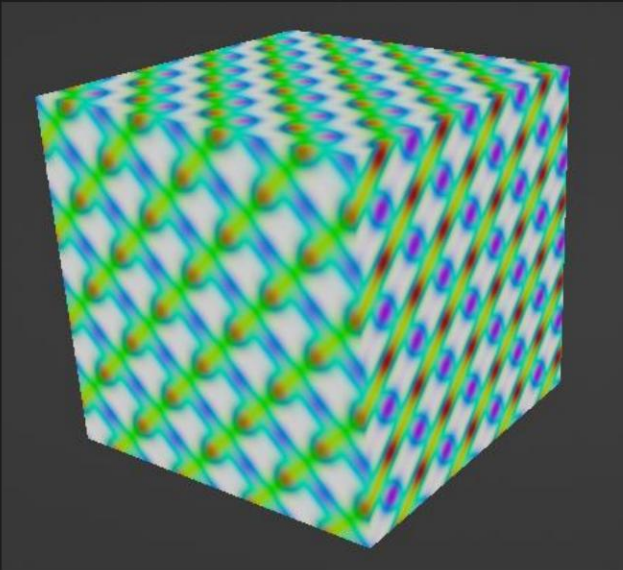
Mincowski ($e=10$)

Magic Texture



Psychedelic texture (official manual expression)

It's not used very much, but it's useful for creating regular, non-monotonous patterns.



008 Making materials through images (1)

How to use basic image textures



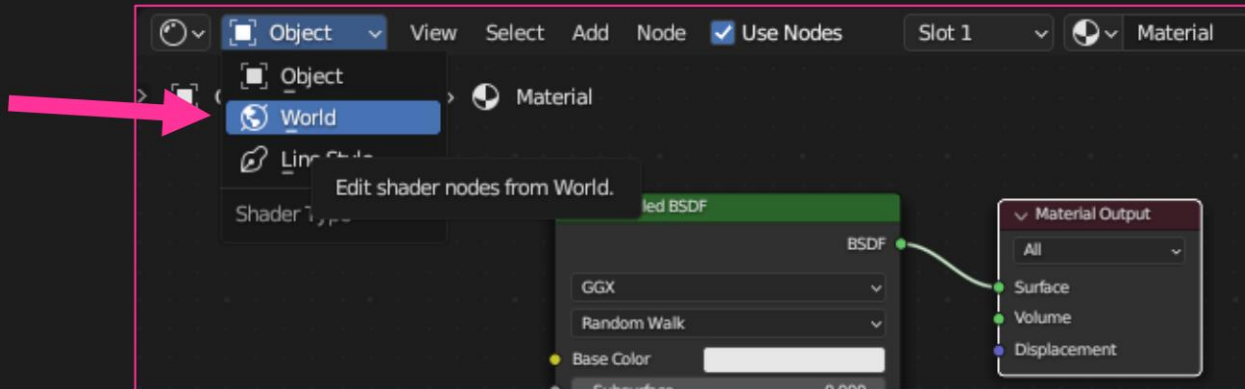
free textures

<https://ambientcg.com/>

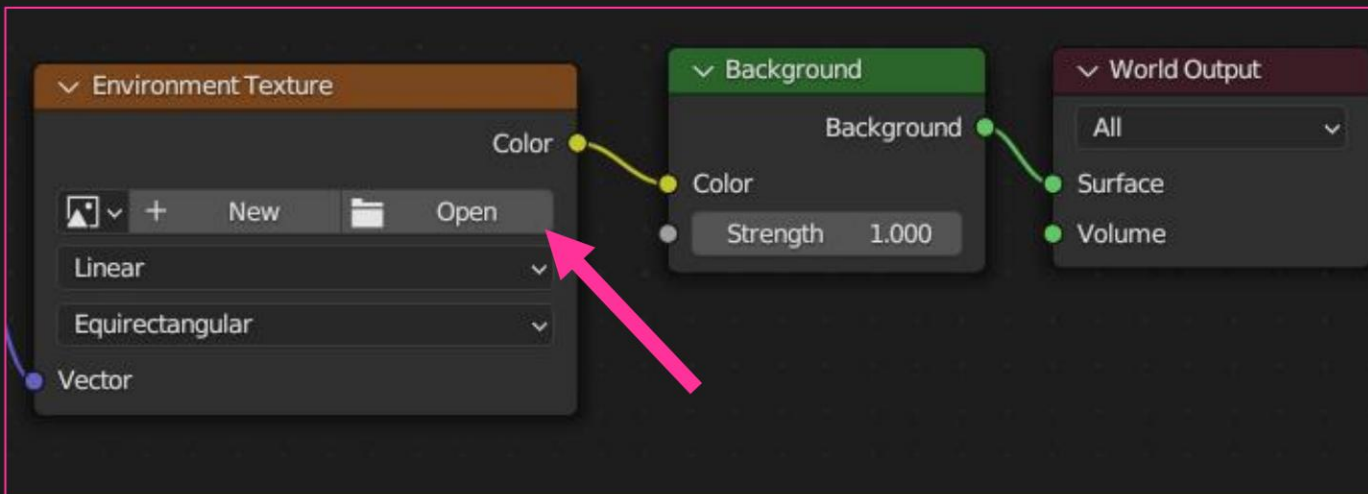
<https://www.3dassets.one/>

background

You can set the background by changing Object to World at the top left of the shader editor.



Click the Background node, and press Ctrl+T to create an image texture node, and an Environment Texture is created. If you open the background image here, it will be applied.

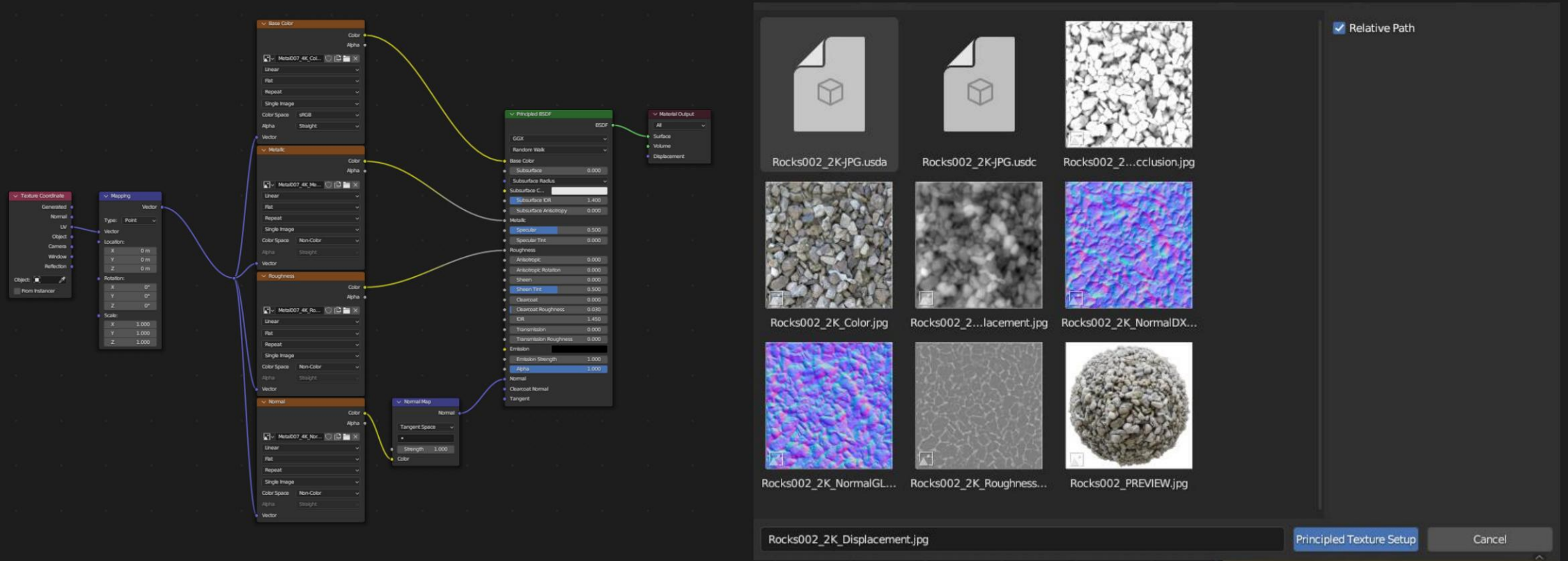


Downloaded Material

Just import the Base Color, Metallic, Roughness, and Normal textures and insert them according to the names.

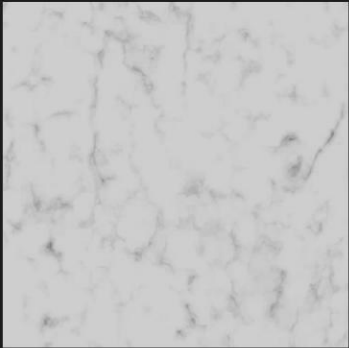
-Normal connects via Normal Map. -Change everything to Non-Color except for the base color.

If you press **Ctrl+Shift+T** in Principled BsdF, it is convenient to open multiple textures at once and connect them automatically.



Downloaded Material

Use the following as needed.



Ambient Occlusion

裂缝

This texture darkens the corners and crevices of the texture.
If necessary, mix with Base Color. (Using Mix Color Multiply)

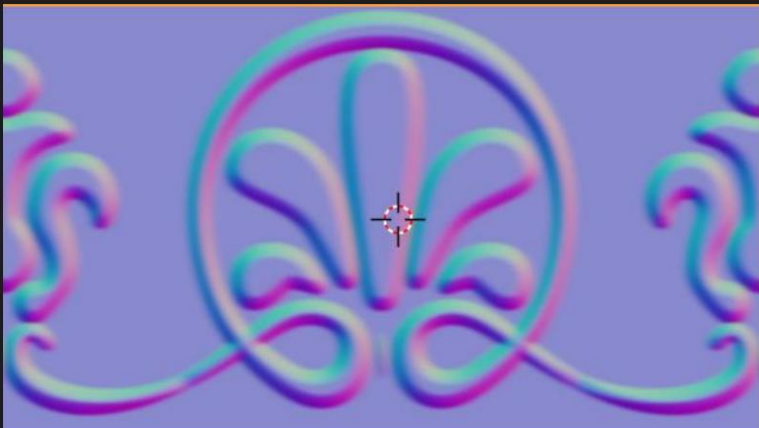
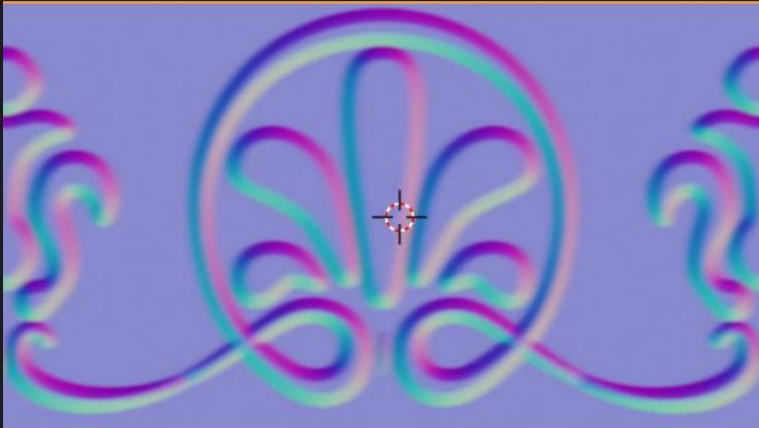


Displacement

This is the texture to create the height.
Connect it to the Height of the Displacement node and then plug it into the Material Output.

Ÿ At this point, there will be no difference before and after application. For details, please refer to Chapter 14.

DirectX vs OpenGL

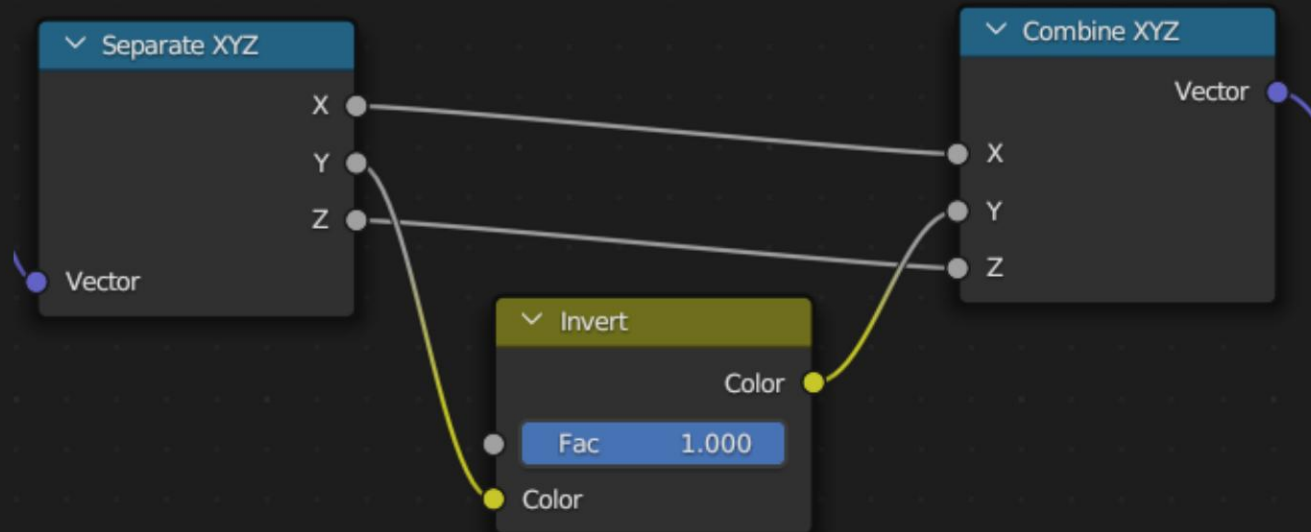


There are two types of normal maps: Direct X and OpenGL. Blender uses the OpenGL method.

不可避免地

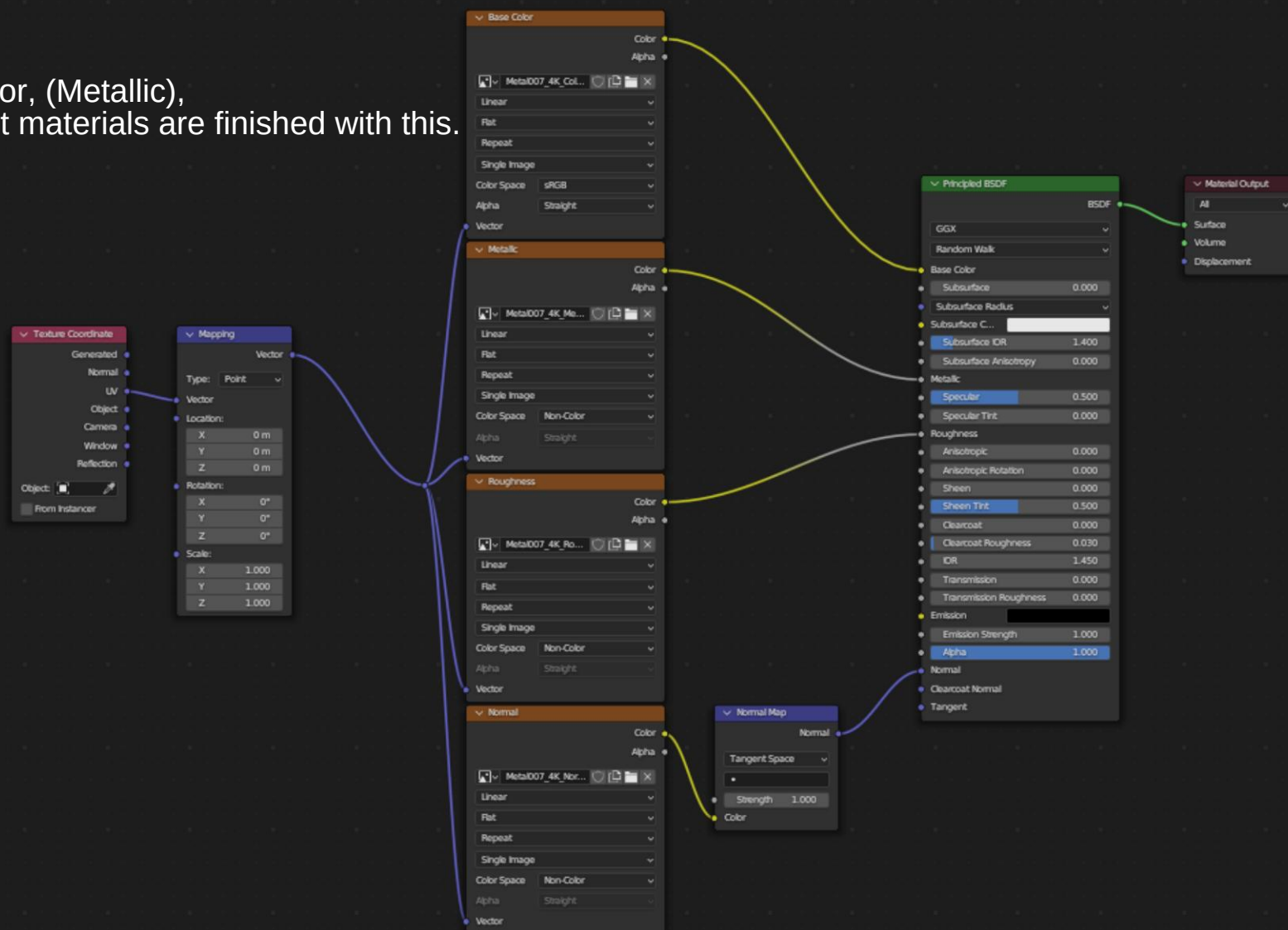
同样在obsidian 笔记中有探索

If you inevitably need to use a normal map created by the Direct X method, you must reverse the Y coordinates as shown below.



Connection(1) Typical Connection

Basic connection is to use Base Color, (Metallic), Roughness, and even Normal . Most materials are finished with this.



Connection (2) additional material settings

Materials that cannot be expressed with the previous four connections require additional settings.

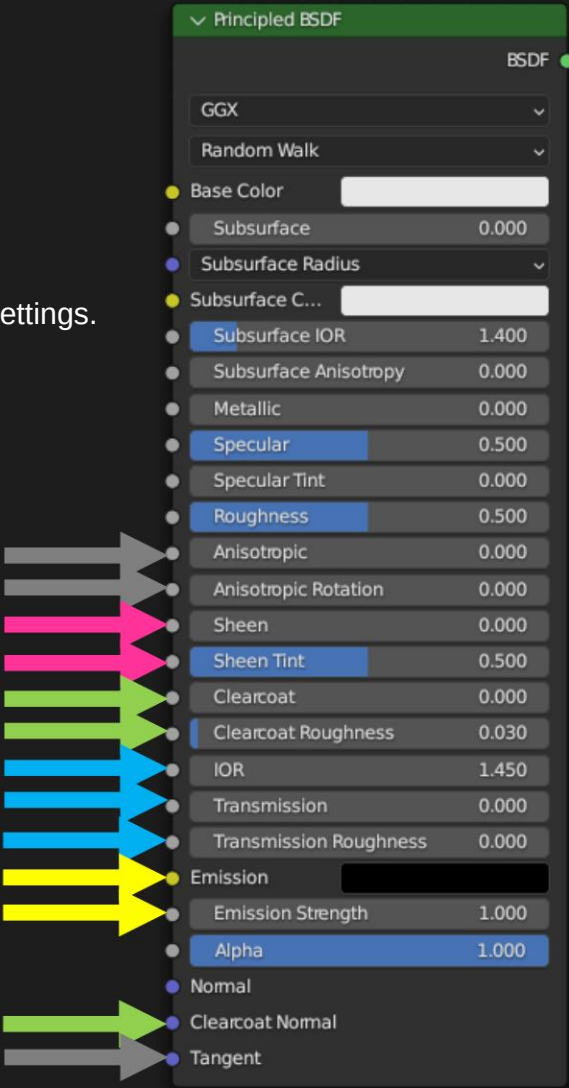
Soft cloth material: Raise the Sheen slider.

Surface coated material: Clearcoat

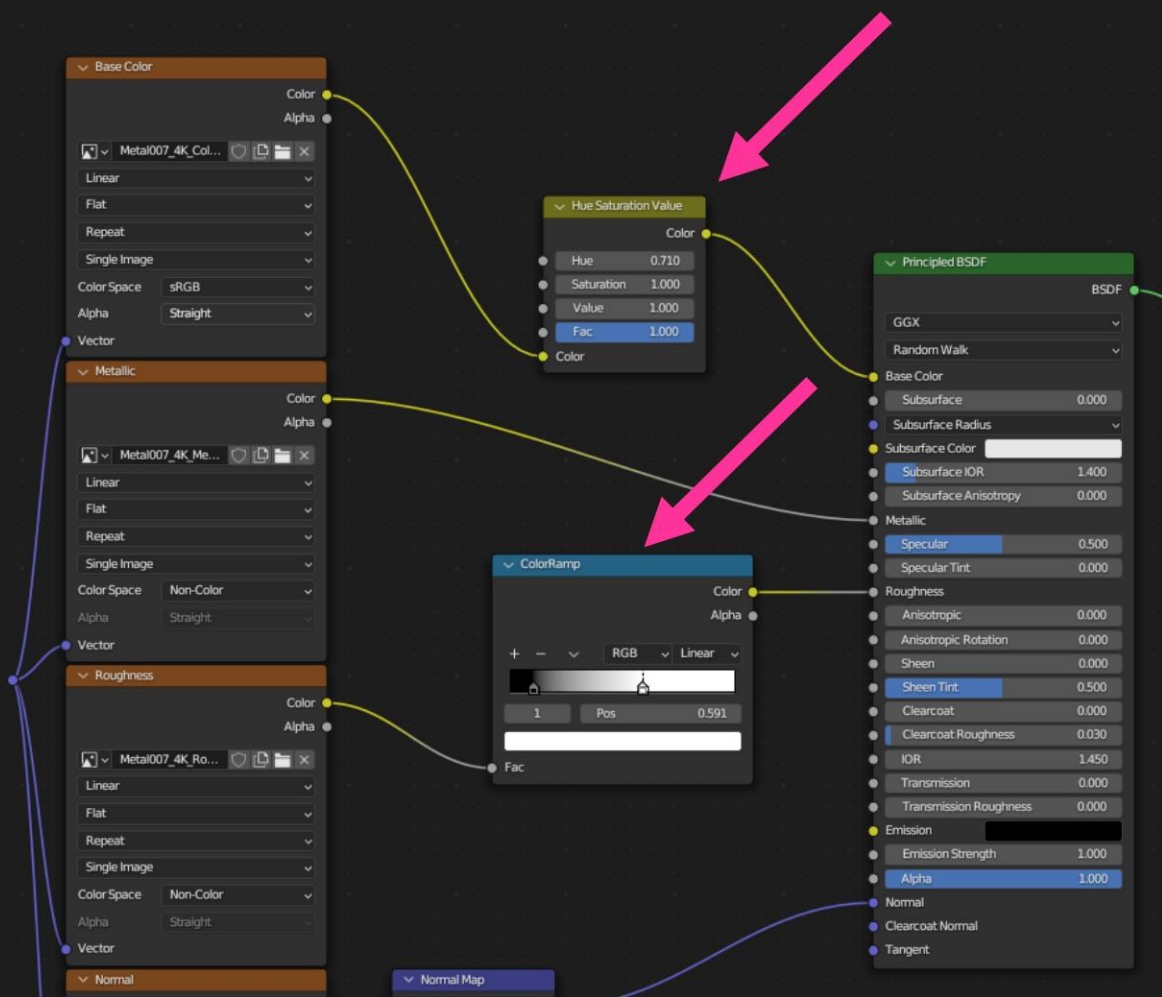
Glass: Transmission

Lighting that emits light: Emission

Brushed Metal : Anisotropic



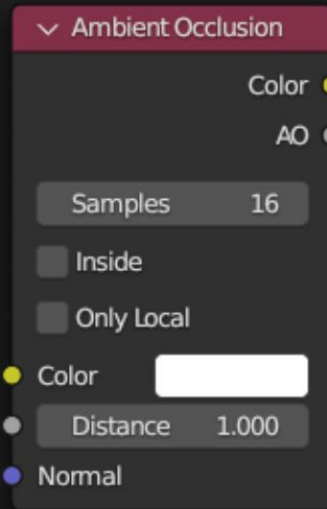
Connection (3) texture correction and additional effects



Change the color of the texture or correct the texture to accentuate the roughness in particular.

You can also use Ambient Occlusion, Bevel to emphasize corners or outlines.

Ambient Occlusion, Bevel

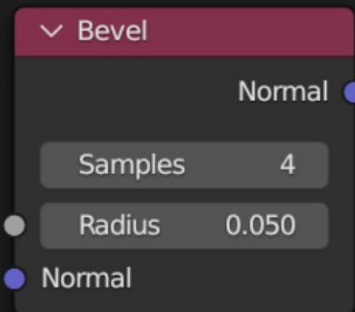


A node that creates Ambient Occlusion based on geometry.

Inside: The function works backwards! In other words, darken the raised parts.

Only Local: Calculates only within a single object. This option only works in Cycles.

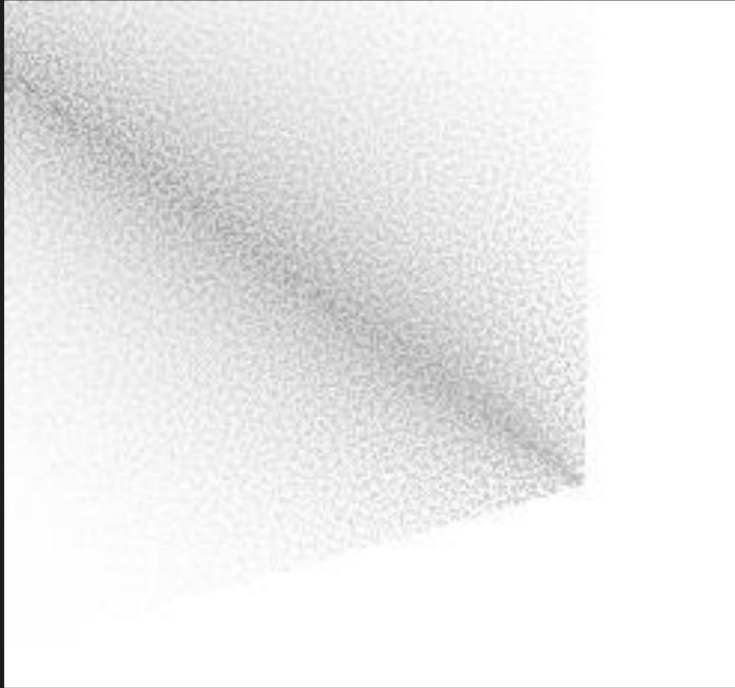
Distance: Specifies the range of the ambient occlusion effect.



Bevel: Gives a kind of blur effect to the input normal to soften the corners. This node only works with Cycles.

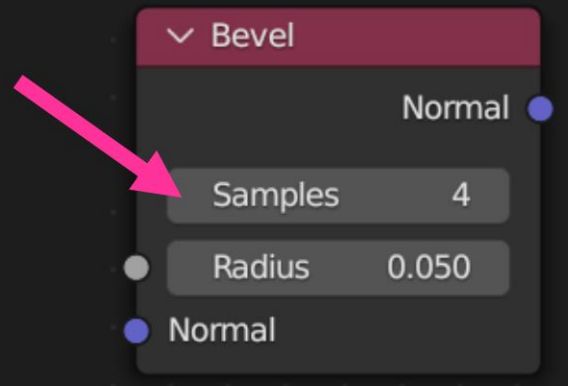
Radius: Controls the strength of the blur effect.

Things to keep in mind when using Bevel and Ambient Occlusion



AO and Bevel are created by computer repeating calculations. Samples control this. Fewer operations produce coarser results. More calculations are more accurate and smoother, but increase rendering time.

The AO and Bevel Samples values are different from the Samples in the Rendering Options. (Of course, this is also affected by rendering samples.)



Connections (4) You can also connect only some of them according to your needs.

BaseColor may not need a texture if the surface has been painted with a solid color, such as paint.

These surfaces can be effectively expressed by connecting only Roughness or Normal.

